



Anomaly Detection and Cyber Attacks Mitigation: Part 2

Dr. Alfán Presekál, Dr. Alex Stefanov

EE4410 - Cyber Security of Power Grids

Faculty of Electrical Engineering, Mathematics and Computer Science, **TU Delft**

Lecture 8 on June 10, 2025

Disclaimer & Ethics

1. **All information provided in the course is for educational purposes only.** The lecturer(s) and teaching assistant(s) are not responsible for any misuse of the information.
2. The course does not promote and is strongly against illegal activities such as unethical hacking, cracking and piracy.
3. **Do not attempt to violate the law with anything contained in the course.** Do not abuse the material provided and be responsible.
4. By participating in the course, **you (student) must accept that you will only lawfully use the material and knowledge you got from this course.**
5. Any actions and / or activities related to the material contained within this course are solely your (student) responsibility. **The misuse of the information in this course can result in criminal charges brought against the persons in question.**
6. The lecturer(s) and teaching assistant(s) will not be held responsible in the event any criminal charges are brought against any individuals misusing the information from this course to break the law.

Outline

1. Graph Convolutional Long Short-Term Memory
2. Time Series Classification with Convolutional Neural Network
3. Anomaly Detection Strategies
4. Attack Graph Model
5. Lab 5

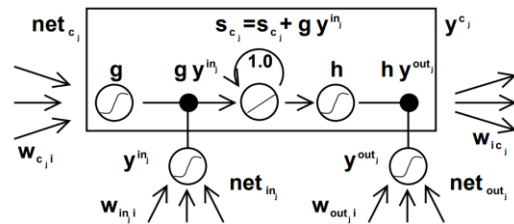


Graph Convolutional Long Short-Term Memory

The AI Models

LSTM (1997)

S. Hochreiter



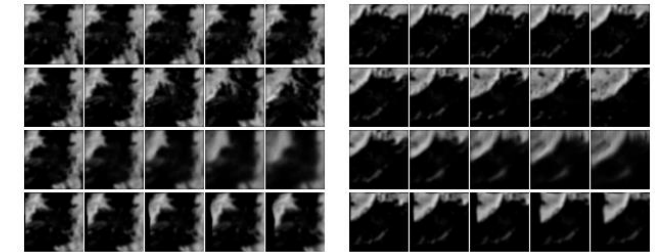
Imagenet /CNN (2012)

Geoffrey Hinton



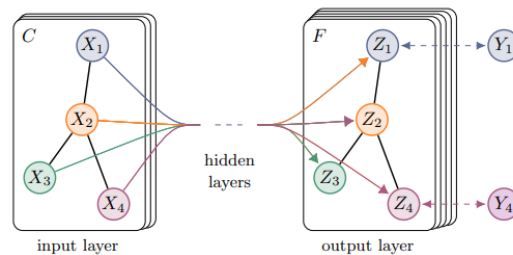
Conv. LSTM (2015)

X. Shi



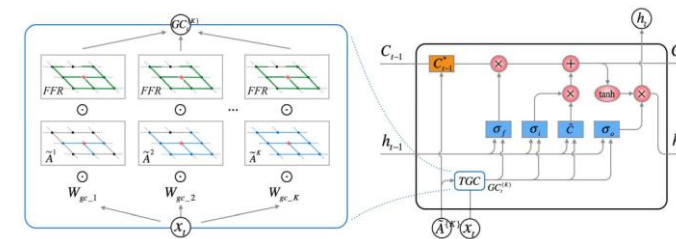
Graph Conv. Net. (2016)

Thomas N. Kipf



Graph Conv. RNN (2020)

Z. Cui

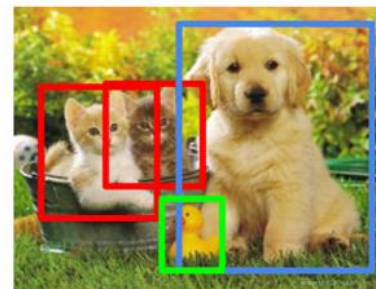


Convolutional Neural Network

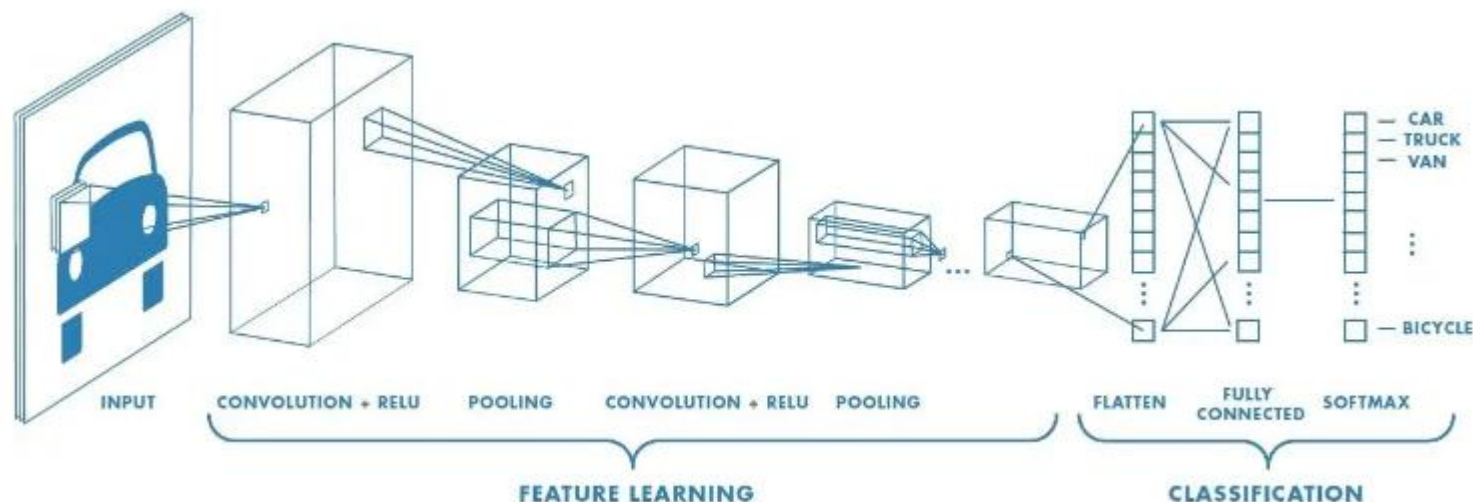
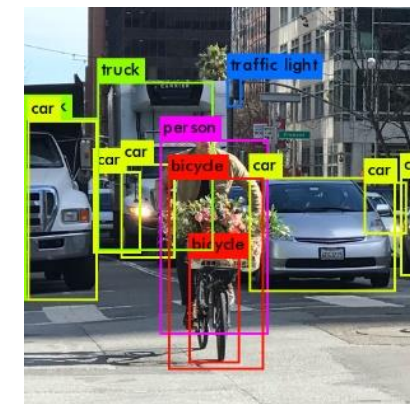


(16, 16)

[	0	0	0	1	2	3	4	5	6	7	8	9	10	11	12	13]
[	13	14	13	15	16	17	18	19	20	21	22	23	24	25	26	13]
[	0	0	13	27	28	29	30	31	32	33	34	35	36	37	14	13]
[	38	13	39	40	41	42	43	44	45	46	47	48	49	50	51	13]
[	14	13	52	53	54	55	56	57	58	59	60	61	62	63	64	65]
[	14	13	66	67	68	69	70	71	72	73	74	75	76	77	78	13]
[	79	80	14	81	82	83	84	85	86	87	88	89	90	91	92	93]
[	0	0	13	94	95	96	97	98	99	100	101	102	103	104	79	0]
[	80	1	105	106	107	108	109	110	111	112	113	114	115	116	1	80]
[	13	117	118	119	120	121	122	123	124	125	126	127	128	129	130	13]
[	13	131	132	133	134	135	136	137	138	139	140	141	142	143	144	13]
[	13	145	146	147	148	149	150	151	152	153	154	155	156	157	158	13]
[	13	159	160	161	162	163	164	165	166	167	168	169	170	171	172	13]
[	0	92	93	173	174	175	176	177	178	179	180	181	182	13	0	0]
[	0	13	183	184	185	186	187	188	189	190	191	192	193	194	13	0]
[	13	195	184	196	197	198	199	200	201	202	203	204	205	206	207	13]]



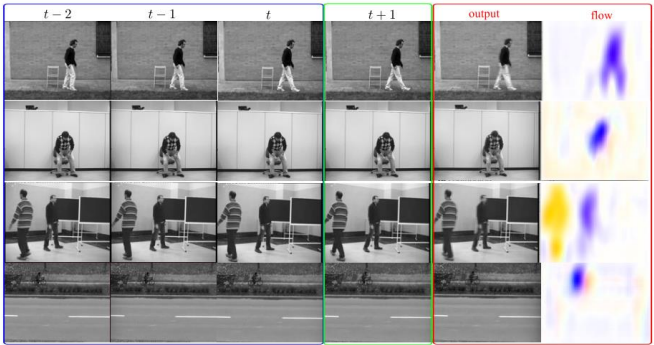
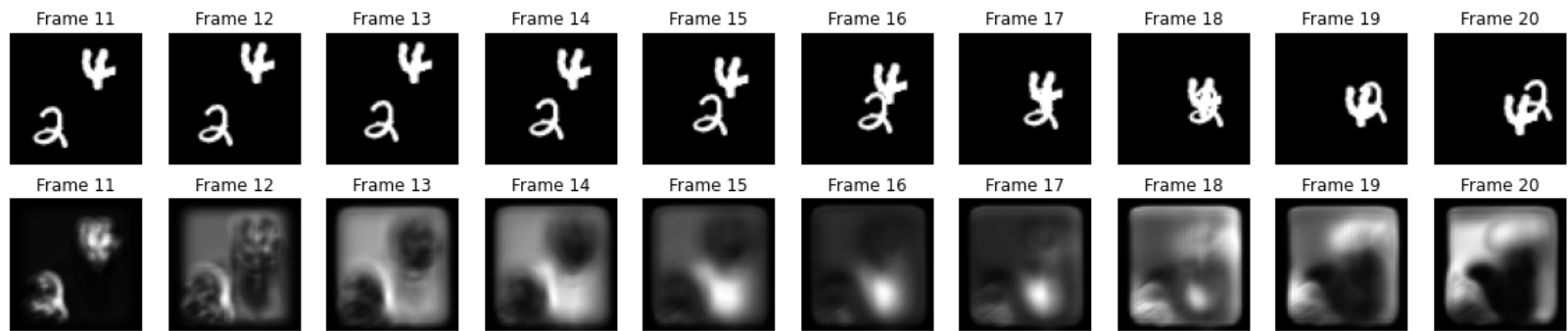
CAT, DOG, DUCK



<https://hannibunny.github.io/mlbook/neuralnetworks/convolutionDemos.html>

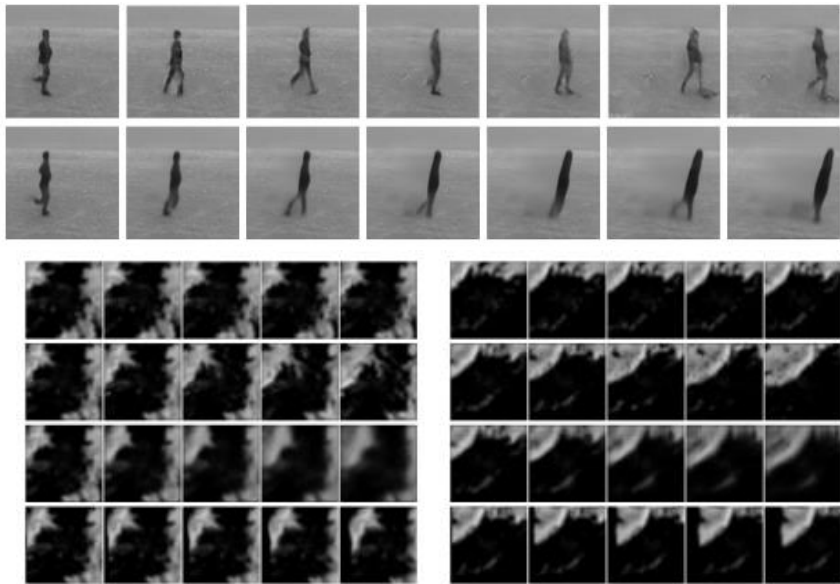
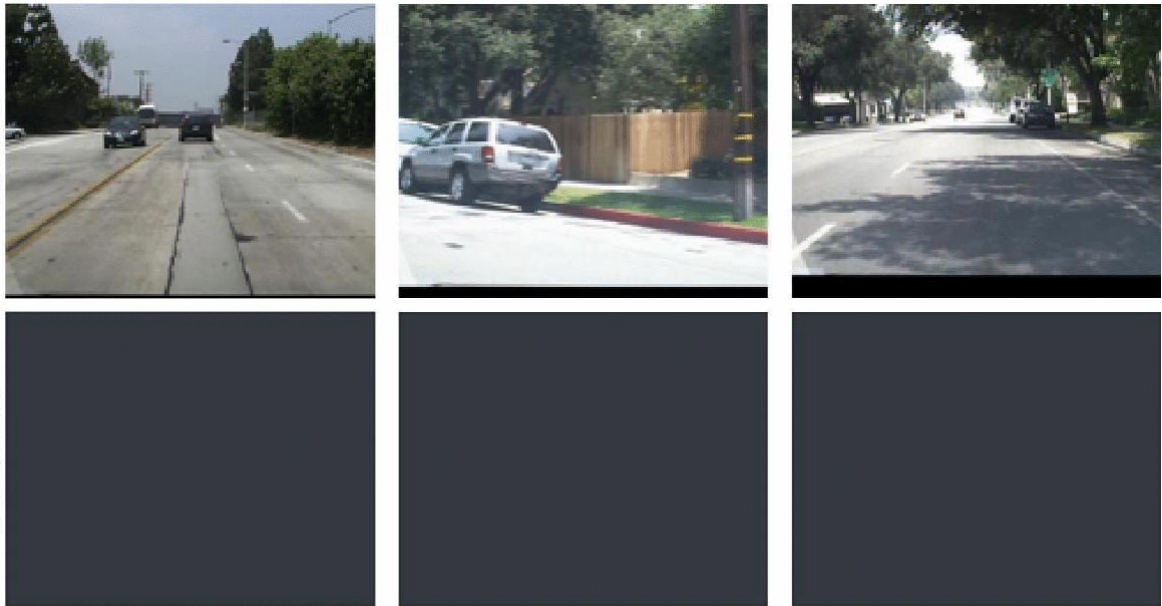
<https://animatedai.github.io/>

Convolutional LSTM



Actual

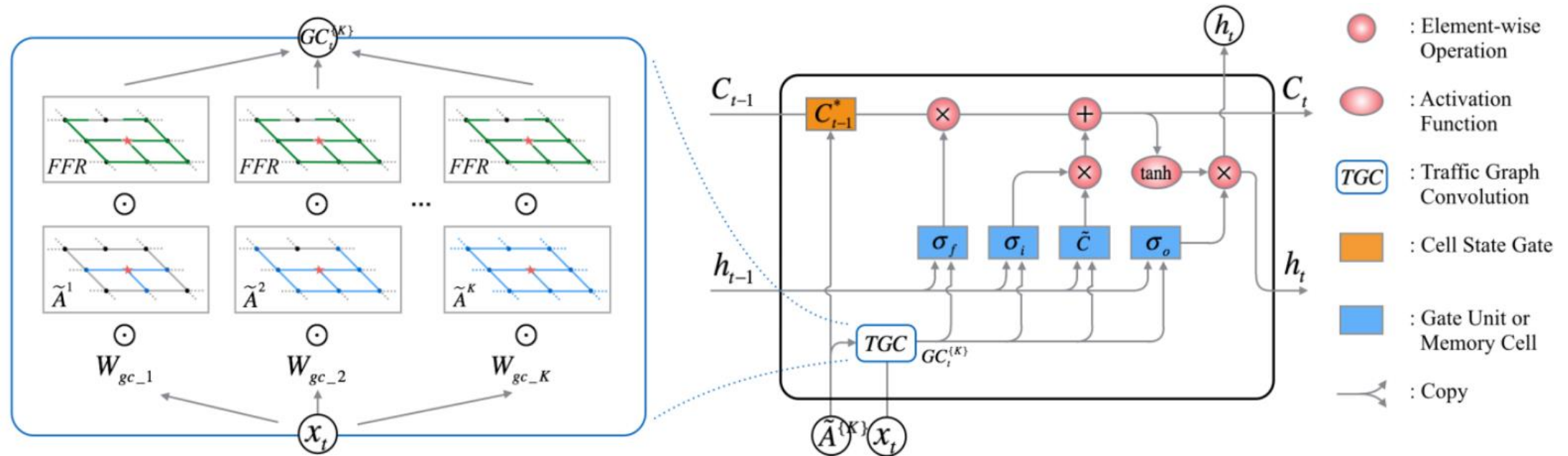
Predicted



Graph Convolutional Long Short-Term Memory (GC-LSTM)

- Graph Convolutional Long Short-Term Memory (GC-LSTM) is a neural network model that integrates the strengths of Graph Convolutional Networks (GCNs) and Long Short-Term Memory (LSTM).
- **Graph Structure Learning:** It utilizes GCNs for their ability to learn and represent graph structures, making it effective in understanding relationships and interactions within networked data.
- **Temporal Dynamics Handling:** Incorporates LSTMs to capture and analyze temporal or sequential data, enabling the model to process time-series information effectively.
- **Advantages:** Offers improved prediction accuracy for time-series data on graphs, outperforming traditional models that don't account for the complex interplay of graph structure and temporal dynamics.
- **Applications:** Primarily used in areas like traffic forecasting, social network analysis, and any domain requiring analysis of time-series data in graph structures.

Graph Convolutional LSTM



Graph Convolutional Layer

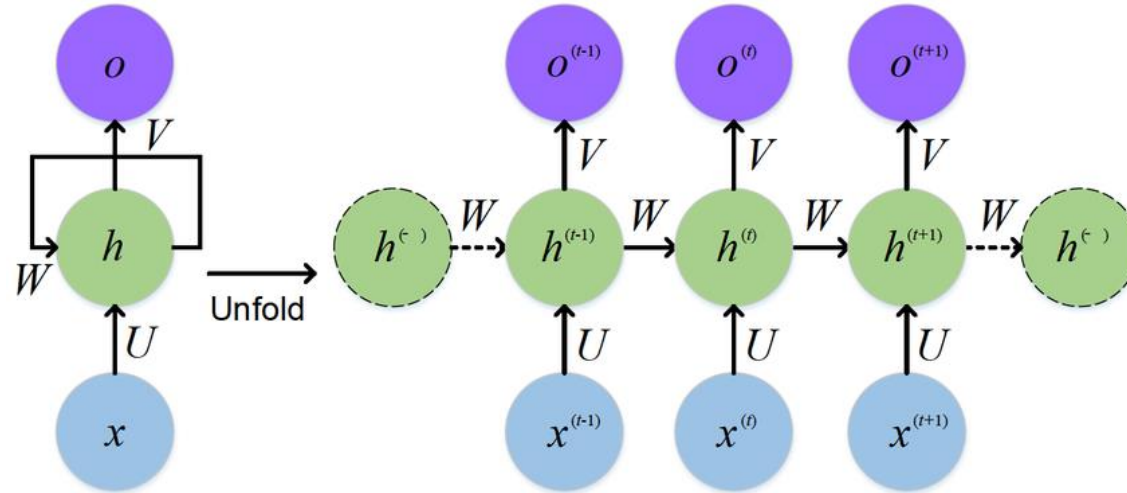
- The graph convolutional layer first captures the spatial structure of the data by aggregating feature information from neighboring nodes in the graph. This process is crucial for learning spatial dependencies within the graph.

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right)$$

- $H^{(l+1)}$ is the matrix of node features at layer $l+1$
- $\tilde{A}$ is the adjacency matrix of the graph with added self-connections
- $\tilde{D}$ is the degree matrix of $\tilde{A}$.
- $W^{(l)}$ are the trainable weights
- σ is the activation function

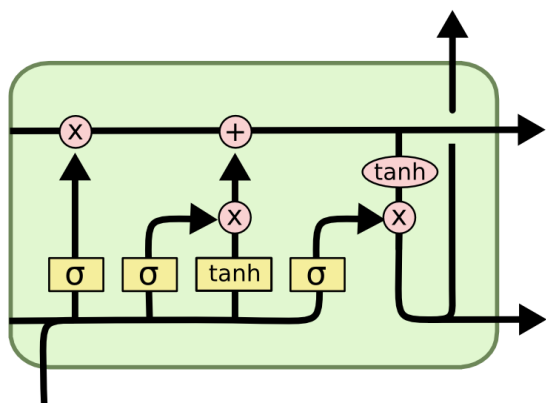
Recurrent Neural Networks

- Recurrent Neural Network (RNN) is a type of artificial neural network designed to recognize patterns in sequences of data.
- **Challenge Vanishing and Exploding Gradient Problem:** Difficulty in learning long-range dependencies due to gradients tending to vanish (become very small) or explode (become very large) as they are propagated back through each time step.

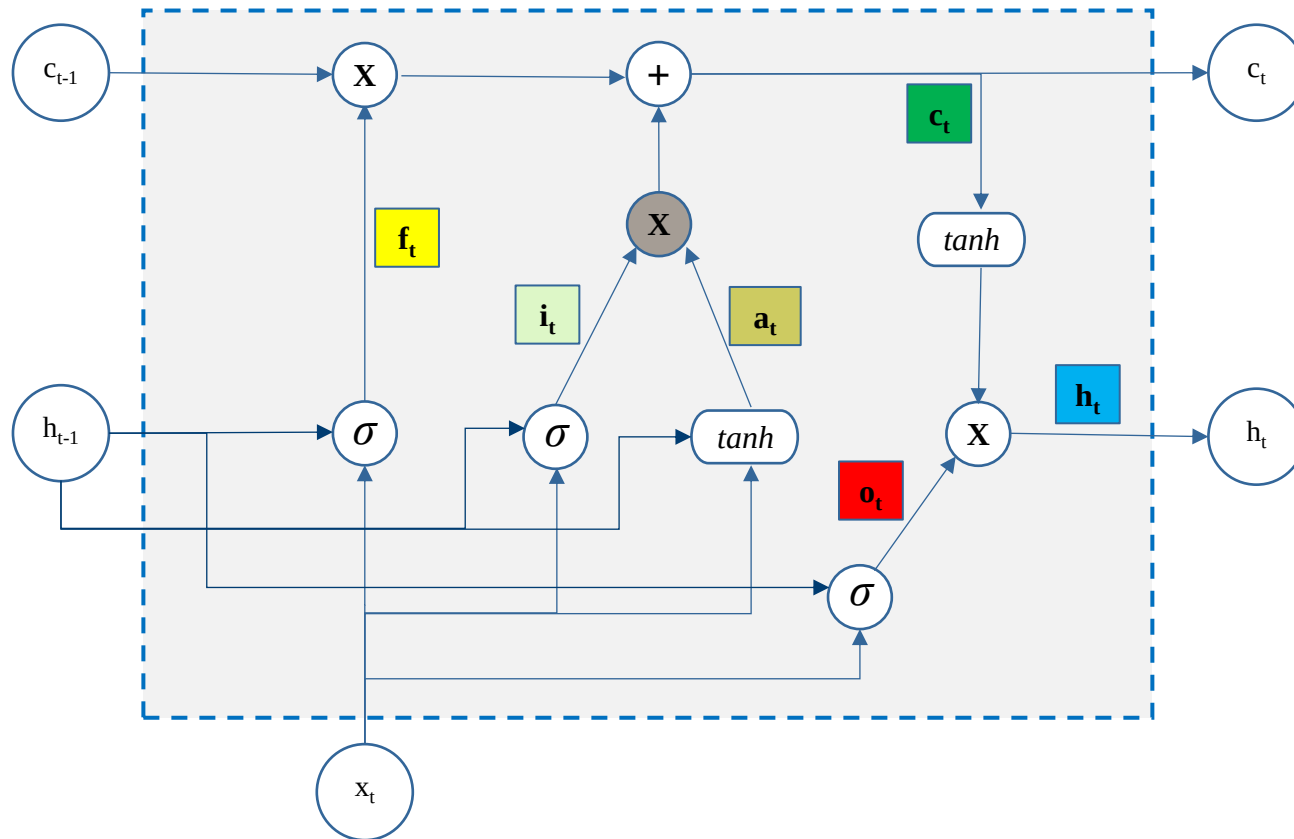


Long Short-Term Memory Layer

- Long Short-Term Memory (LSTM) is a variant of RNN.
- LSTM architecture was specifically created to address the vanishing gradient problem.
- Uses a combination of hidden units, elementwise products, and sums between units to implement gates that control “memory cells.” Memory cells are designed to retain information without modification for long periods of time.
- LSTM has its own input and output gates, which are controlled by learnable weights that are a function of the current observation and the hidden units at the previous time step.



LSTM Cell Architecture



f_t : forget gate
 i_t : input gate
 o_t : output gate
 c_t : cell state / long term memory
 a_t : input activation function
 h_t : hidden state

$$f_t = \sigma((W_f \otimes x_t) + (U_f \otimes h_{t-1}) + b_f)$$

$$i_t = \sigma((W_i \otimes x_t) + (U_i \otimes h_{t-1}) + b_i)$$

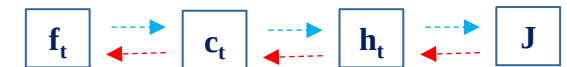
$$o_t = \sigma((W_o \otimes x_t) + (U_o \otimes h_{t-1}) + b_o)$$

$$a_t = \tanh((W_a \otimes x_t) + (U_a \otimes h_{t-1}) + b_a)$$

$$c_t = (f_t \otimes c_{t-1}) + (i_t \otimes a_t)$$

$$h_t = o_t \otimes \tanh(c_t)$$

$$\text{Gates} = \begin{bmatrix} f_t \\ i_t \\ a_t \\ o_t \end{bmatrix} \quad W = \begin{bmatrix} W_f \\ W_i \\ W_a \\ W_o \end{bmatrix} \quad U = \begin{bmatrix} U_f \\ U_i \\ U_a \\ U_o \end{bmatrix} \quad b = \begin{bmatrix} b_f \\ b_i \\ b_a \\ b_o \end{bmatrix}$$



Stacked LSTM

$$f_t = \sigma((W_f \times x_t) + (U_f \times h_{t-1}) + b_f)$$

$$i_t = \sigma((W_i \times x_t) + (U_i \times h_{t-1}) + b_i)$$

$$o_t = \sigma((W_o \times x_t) + (U_o \times h_{t-1}) + b_o)$$

$$c'_t = \tanh((W_c \times x_t) + (U_c \times h_{t-1}) + b_c)$$

$$c_t = (f_t \cdot c_{t-1}) + (i_t \cdot c'_t)$$

$$h_t = o_t \cdot \tanh(c_t)$$

$$f_{t+1} = \sigma((W_f \times x_{t+1}) + (U_f \times h_t) + b_f)$$

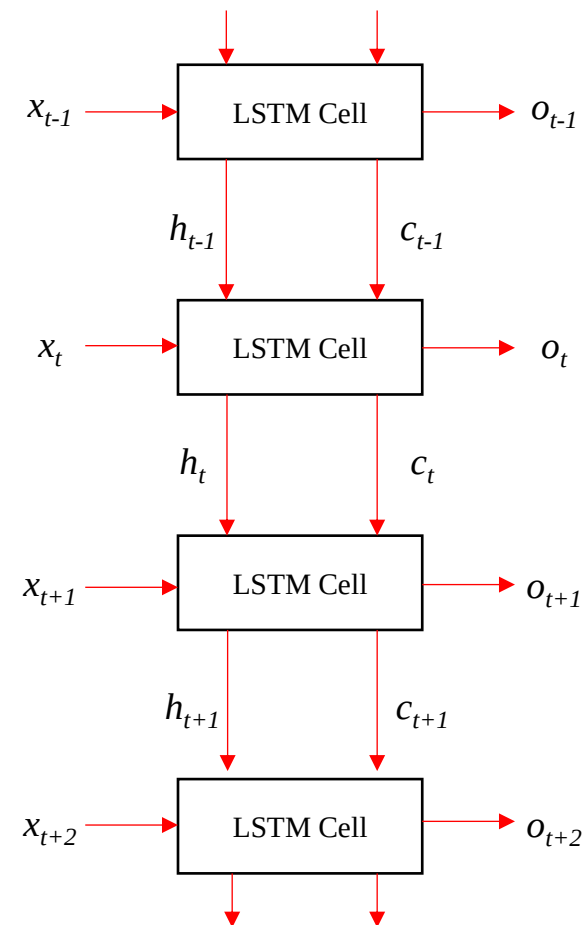
$$i_{t+1} = \sigma((W_i \times x_{t+1}) + (U_i \times h_t) + b_i)$$

$$o_{t+1} = \sigma((W_o \times x_{t+1}) + (U_o \times h_t) + b_o)$$

$$c'_{t+1} = \tanh((W_c \times x_{t+1}) + (U_c \times h_t) + b_c)$$

$$c_{t+1} = (f_{t+1} \cdot c_t) + (i_{t+1} \cdot c'_{t+1})$$

$$h_{t+1} = o_{t+1} \cdot \tanh(c_{t+1})$$



Graph Convolutional Long Short-Term Memory (1)

$$GCN_t^k \leftarrow (W_{gcn} \odot \hat{A}^k) X_t \quad (1)$$

- GCN function is used to obtain the nodal features as described in (1)
- GCN operates based on the Hadamard product multiplication ($\odot$) of the weight matrix (W_{gcn}), adjacency matrix (A), and node features from the observed traffic data (X_t).
- The adjacency matrix captures information related to the OT network topology. The adjacency matrix (A) is added with the identity matrix (I) to form a modified adjacency matrix ($\hat{A}$).
- The equation also considers the number of hops from a communication node to neighboring nodes, i.e., k as an exponent of $\hat{A}$.

Graph Convolutional Long Short-Term Memory (2)

- The LSTM functions and processes inside an LSTM cell are described in (2 - 7).
- There are six main sub-equations in the LSTM process, including the forget gate (f_t), input gate (i_t), output gate (o_t), internal cell state (c'_t), transferable cell state (c_t), and hidden state (h_t).

$$GCN_t^k \leftarrow (W_{gcn} \odot \hat{A}^k) X_t \quad (1)$$

$$f_t = \sigma((W_f GCN_t^k) + (U_f h_{t-1}) + b_f) \quad (2)$$

$$i_t = \sigma((W_i GCN_t^k) + (U_i h_{t-1}) + b_i) \quad (3)$$

$$o_t = \sigma((W_o GCN_t^k) + (U_o h_{t-1}) + b_o) \quad (4)$$

$$c'_t = \tanh((W_c GCN_t^k) + (U_c h_{t-1}) + b_{c'}) \quad (5)$$

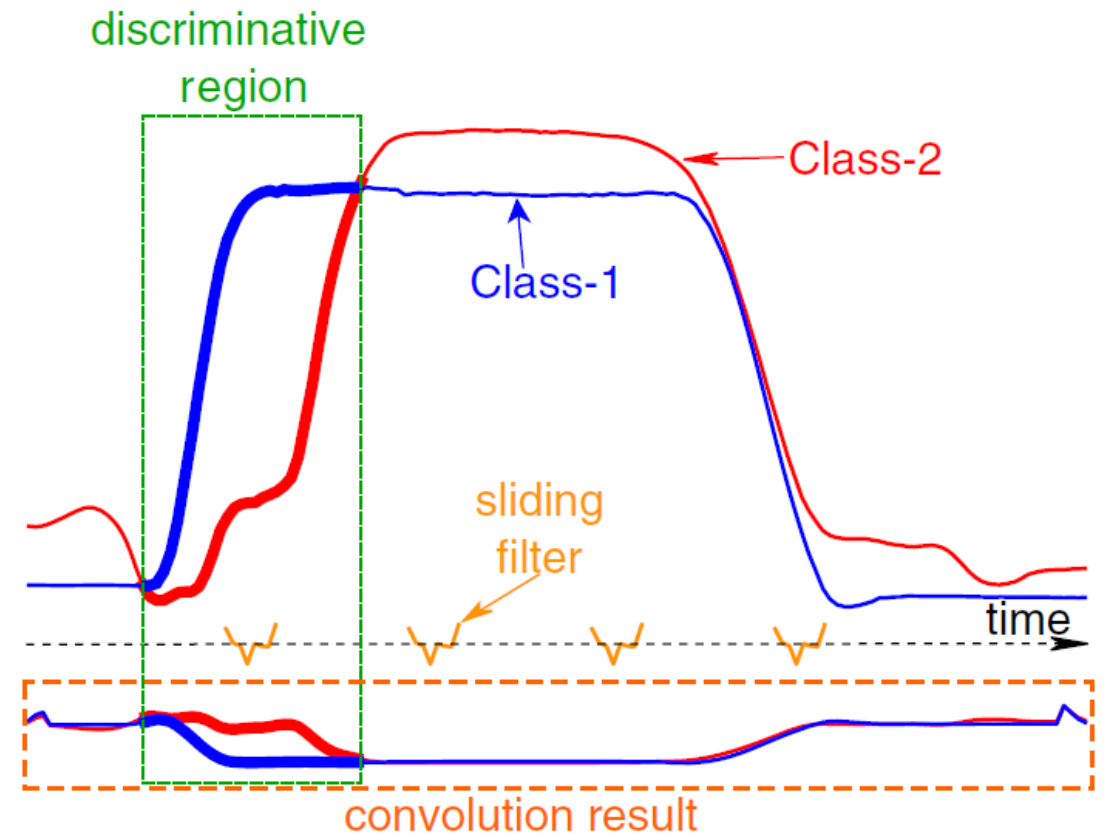
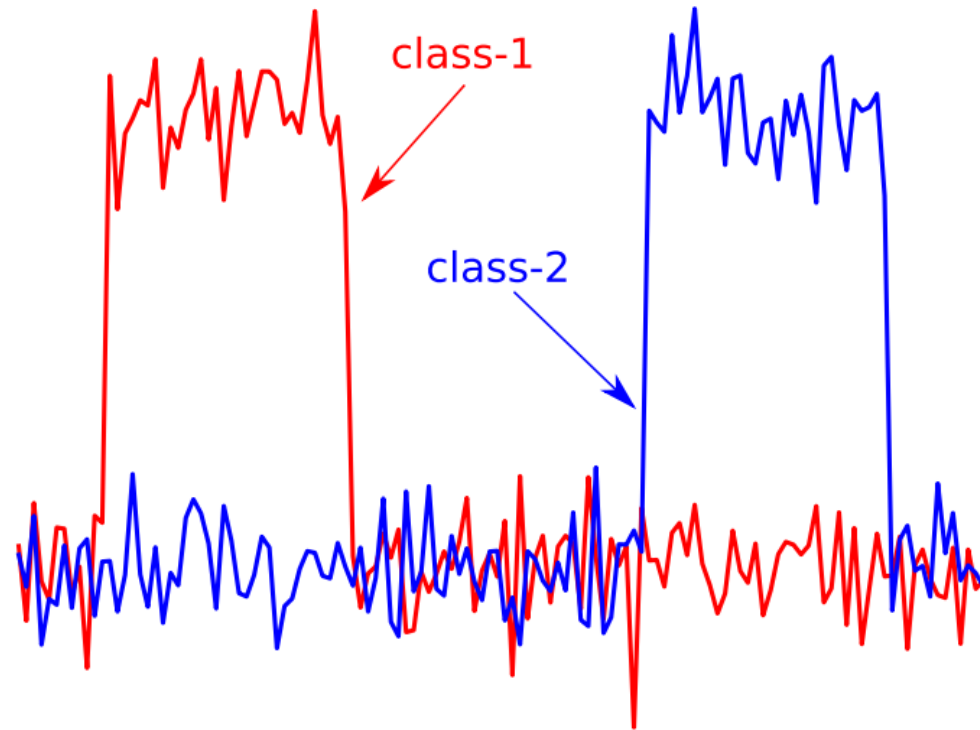
$$c_t = (f_t \odot c_{t-1}) + (i_t \odot c'_t) \quad (6)$$

$$h_t = o_t \odot \tanh(c_t) \quad (7)$$

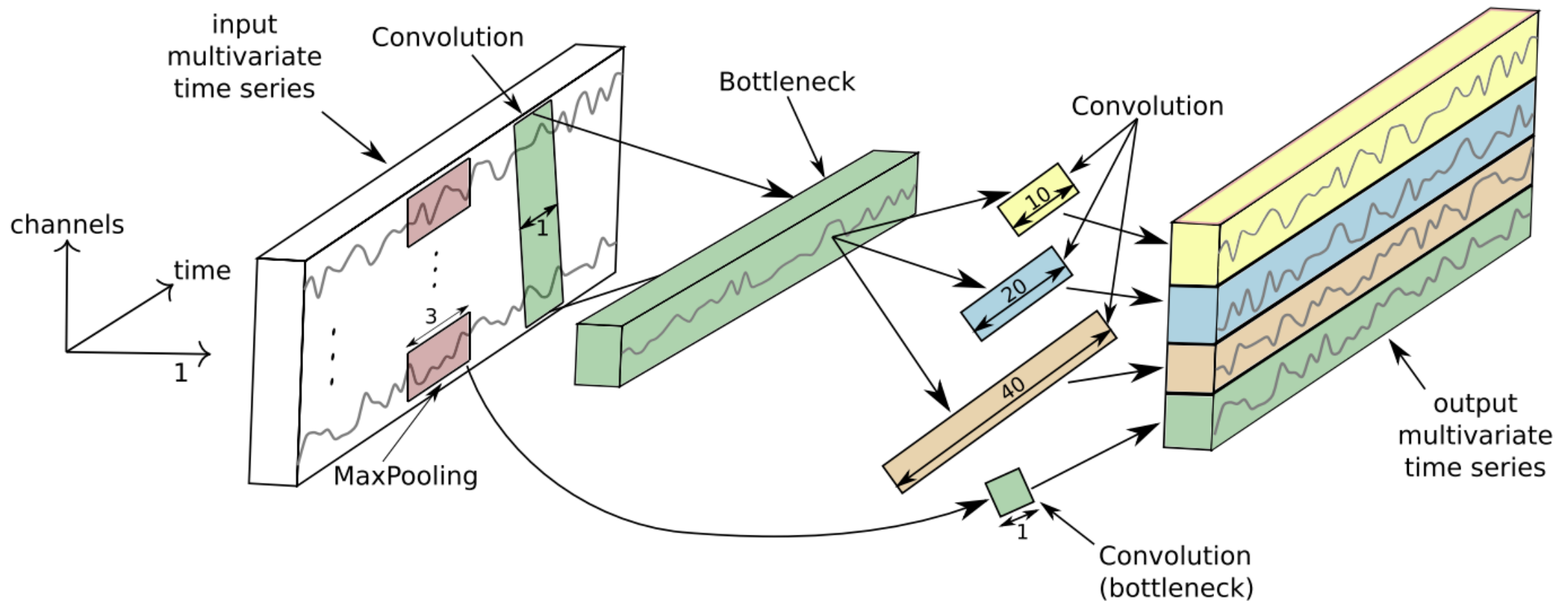


Time Series Classification with Convolutional Neural Networks

Time Series Classification

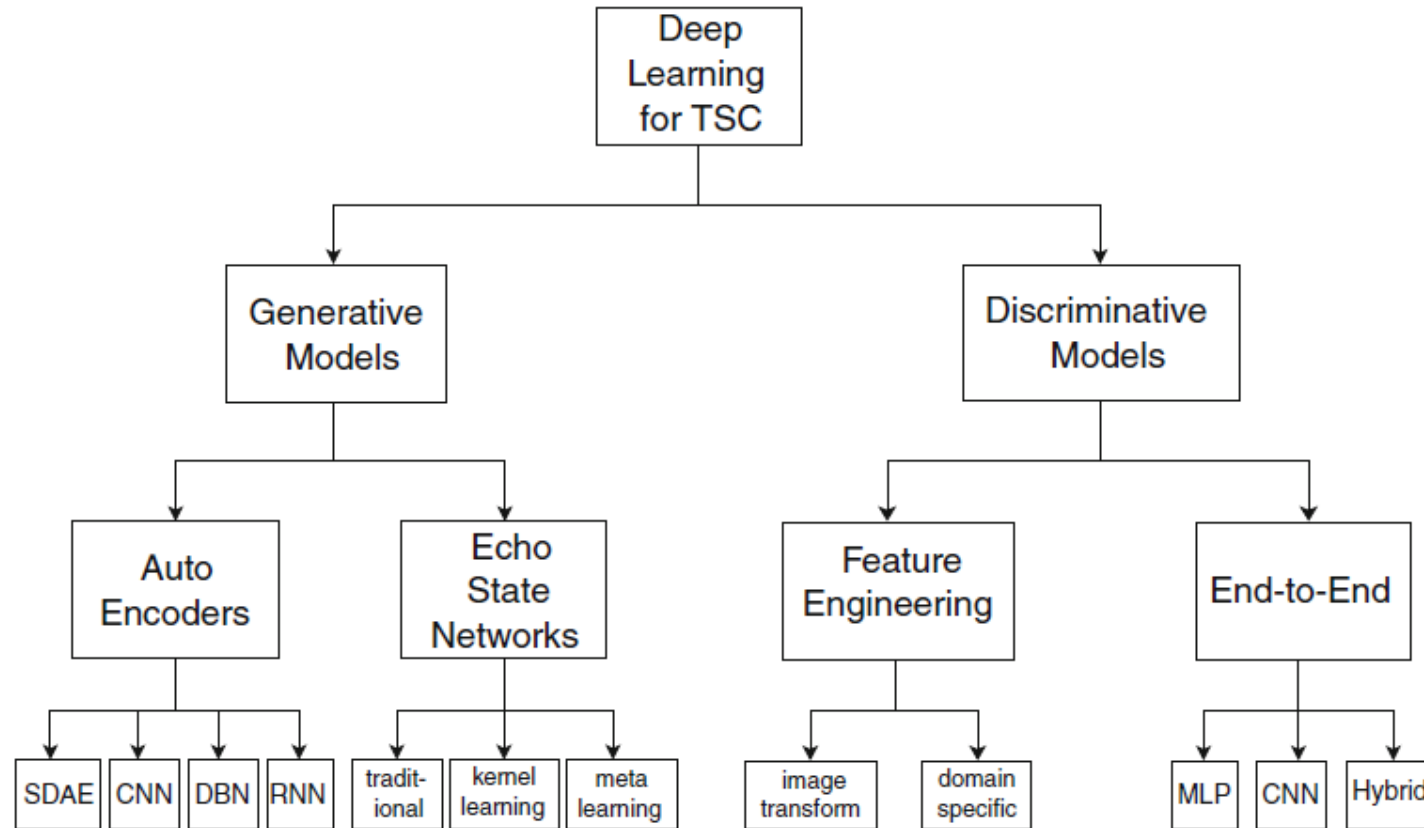


Inception for Time Series Classification



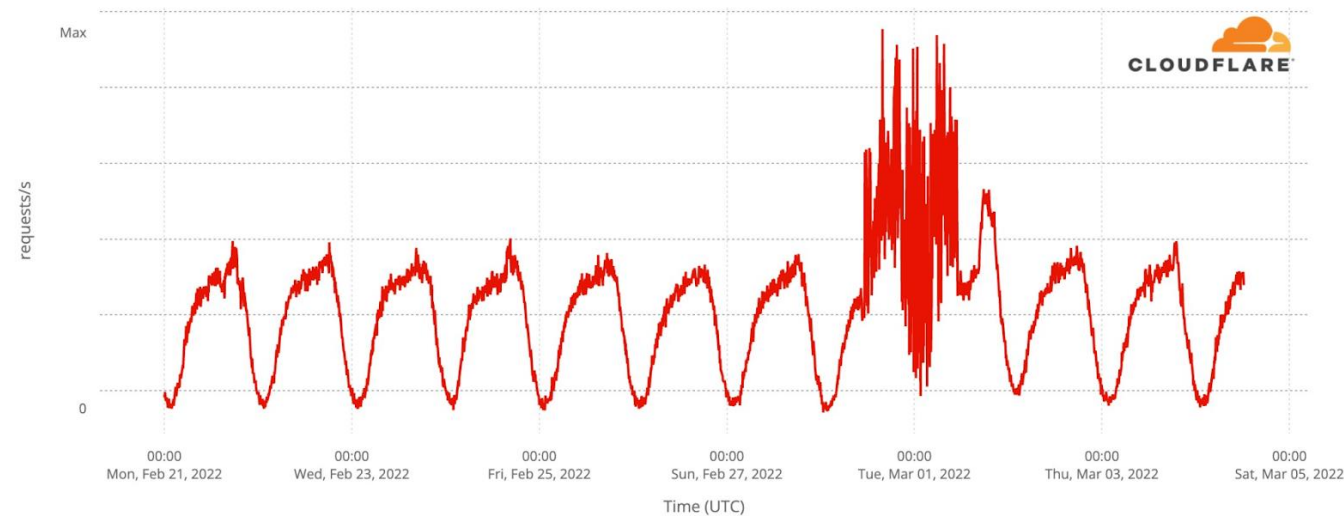
Deep Learning Taxonomy for Time Series Classification

An overview of the different deep learning approaches for time series classification

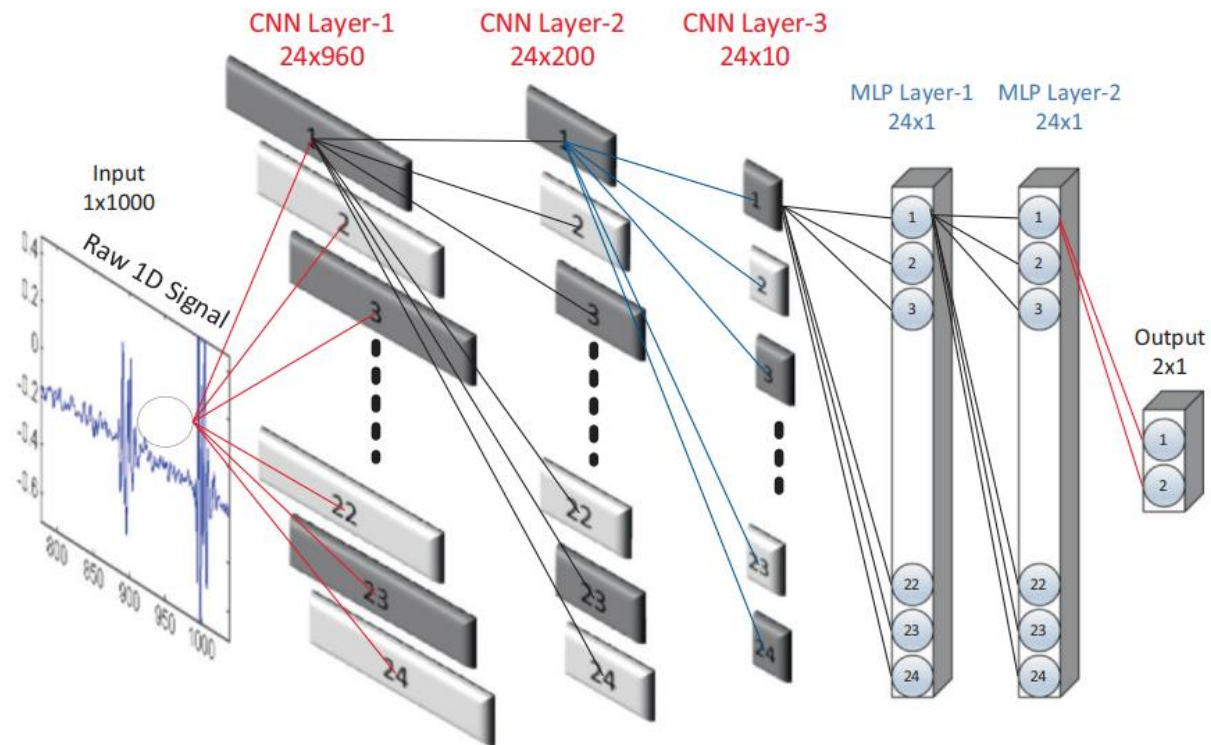


1D CNN

- 2D CNNs are primarily applied for 2D signals such as images and video frames
- 1D CNNs have recently been proposed for 1D data, i.e., signal, biomedical data, anomaly detection, power electronics, etc.
- 1D CNN is simpler than 2D architecture which leads to lower computation cost.



1D CNN



A sample 1D CNN configuration with 3 CNN and 2 MLP layers.

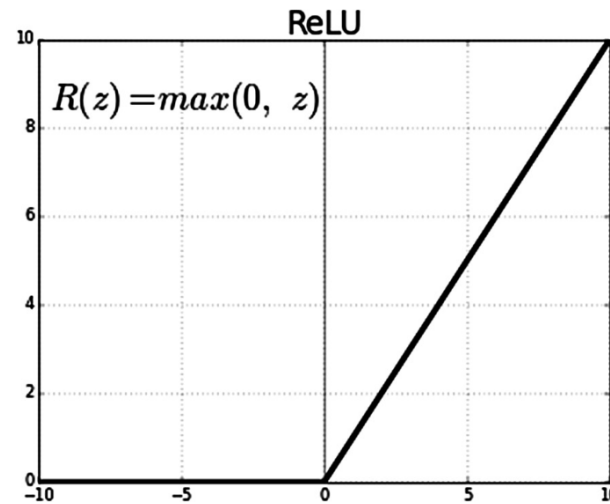
Components of 1D CNN

- Convolutional Layer

$$y(t) = (x * w)(t) = \sum_{s=-a}^a x(s) \cdot w(t - s)$$

- Activation Function Rectified Linear Unit ReLU

$$y = \max(0, x)$$



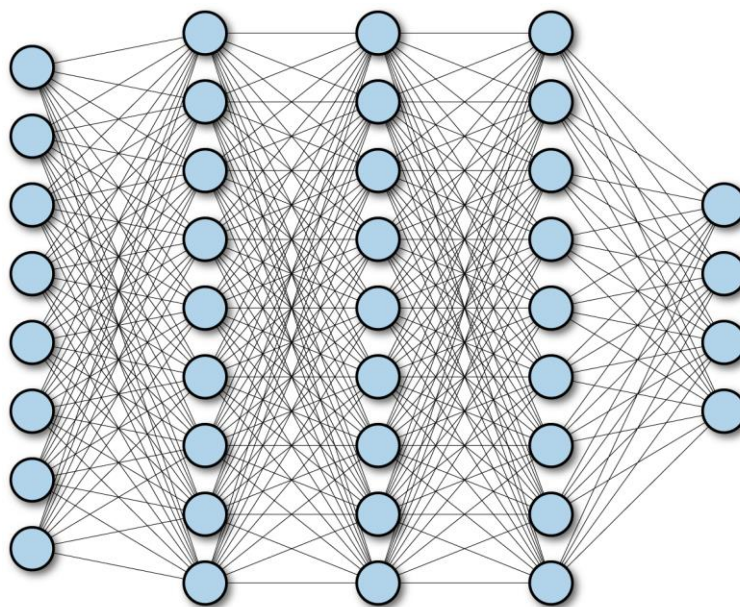
Components of 1D CNN

- Max Pooling Layer

$$y(t) = \max(x(t), x(t+1), \dots, x(t+k))$$

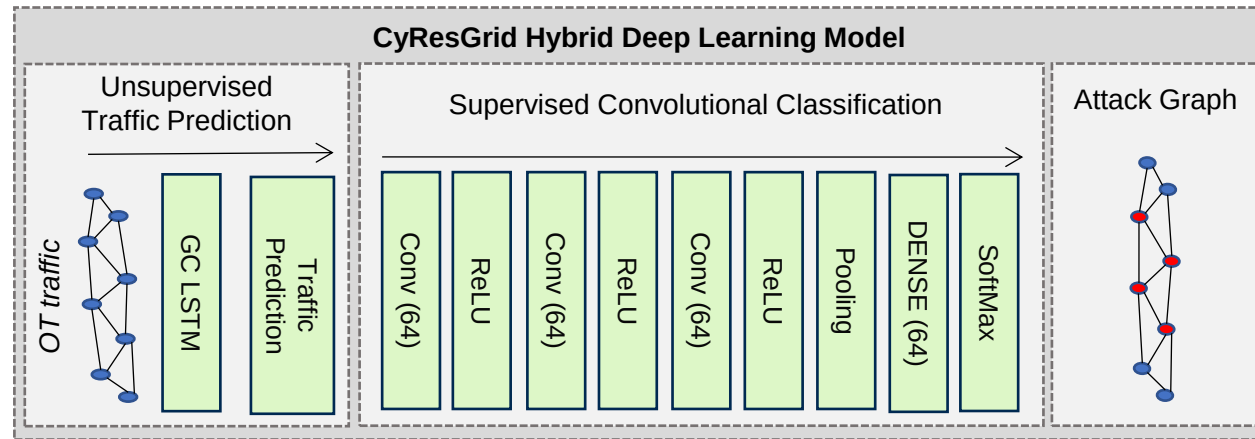
- Fully Connected Dense Layer

$$y = Wx + b$$



CNN Time Series Classification

- The figure shows the architecture of CyResGrid hybrid deep learning model that consists of a GC-LSTM layer, three layers of convolutional neural network, and one layer of fully connected neural network (dense).



- To design the CNN architecture, a hyperparameter tuning is performed using Bayesian optimization.

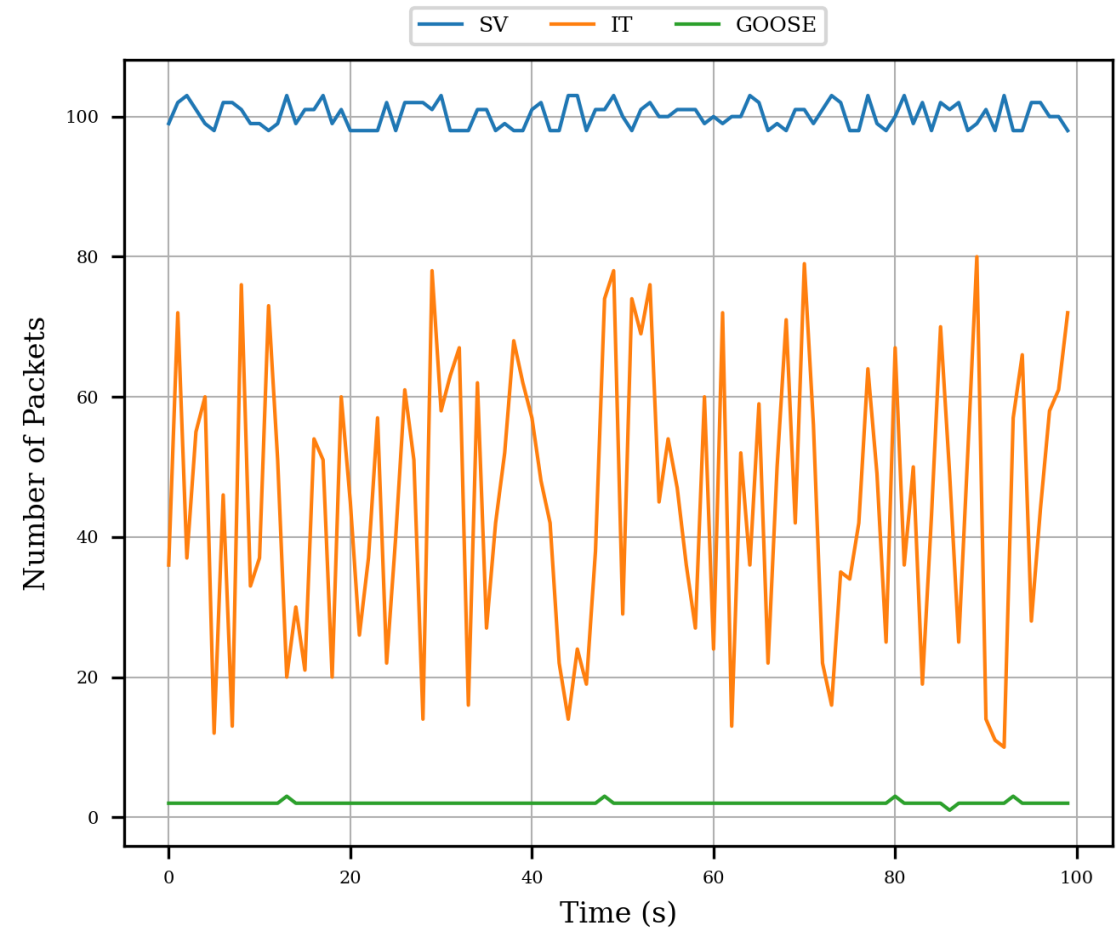
$$x^* = \arg \max_x f(x)$$



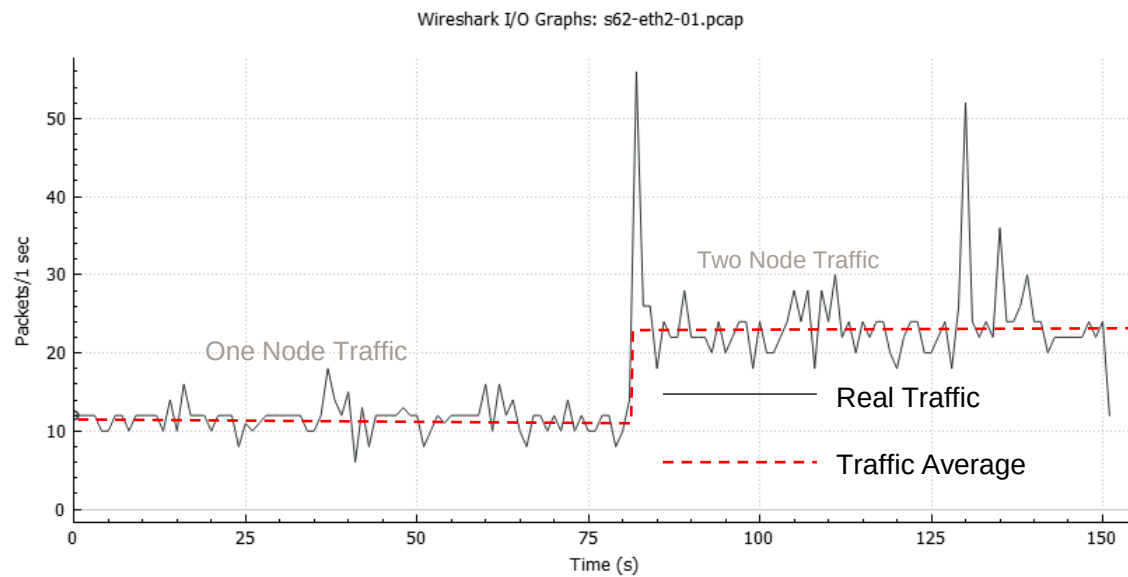
Anomaly Detection Strategies

Operational Technology Traffic Characteristic

- IT and OT traffic characteristics are distinct
- The network traffic in OT systems is generated from automated processes with deterministic and homogenous behavior
- IT system traffic consists of user-generated data with a stochastic behavior
- Figure depict sample of OT traffic (SV and Goose) and IT traffic



OT Time Series Traffic



Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration
100.6.0.11	100.6.0.22	284	25 k	142	15 k	142	9372	0.154013	151.4277
100.6.0.11	100.6.0.23	133	11 k	66	7354	67	4430	82.660644	68.7411
100.6.0.11	100.6.0.21	5	351	2	140	3	211	130.172174	8.3044
100.6.0.12	100.6.0.22	282	25 k	141	15 k	141	9306	0.078381	150.9763
100.6.0.12	100.6.0.23	133	11 k	66	7354	67	4430	82.637832	68.7794
100.6.0.12	100.6.0.21	3	214	1	74	2	140	130.192854	0.0038
100.6.0.13	100.6.0.22	260	25 k	130	17 k	130	8580	0.000000	150.3847
100.6.0.13	100.6.0.23	121	11 k	60	7862	61	4034	82.660685	68.5670
100.6.0.13	100.6.0.21	3	214	1	74	2	140	130.213502	0.0045
100.6.0.14	100.6.0.22	284	25 k	142	15 k	142	9372	0.751143	150.2449
100.6.0.14	100.6.0.23	133	11 k	66	7354	67	4430	82.635440	68.7358
100.6.0.14	100.6.0.21	3	214	1	74	2	140	130.234880	0.0067
100.6.0.15	100.6.0.22	292	23 k	146	14 k	146	9636	0.557494	150.1030
100.6.0.15	100.6.0.23	137	11 k	68	6506	69	4562	82.581120	68.2270
100.6.0.15	100.6.0.21	3	214	1	74	2	140	130.262257	0.0041
100.6.0.16	100.6.0.22	270	25 k	135	16 k	135	8910	0.544806	150.2925
100.6.0.16	100.6.0.23	129	12 k	64	7886	65	4298	82.688348	68.1520
100.6.0.16	100.6.0.21	3	214	1	74	2	140	130.279916	0.0079

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
✓ Packet Lengths	298	88.27	42	132	0.0115	100%	0.0800	12.716
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	155	64.14	42	66	0.0060	52.01%	0.0600	12.716
80-159	143	114.42	96	132	0.0055	47.99%	0.0300	2.216

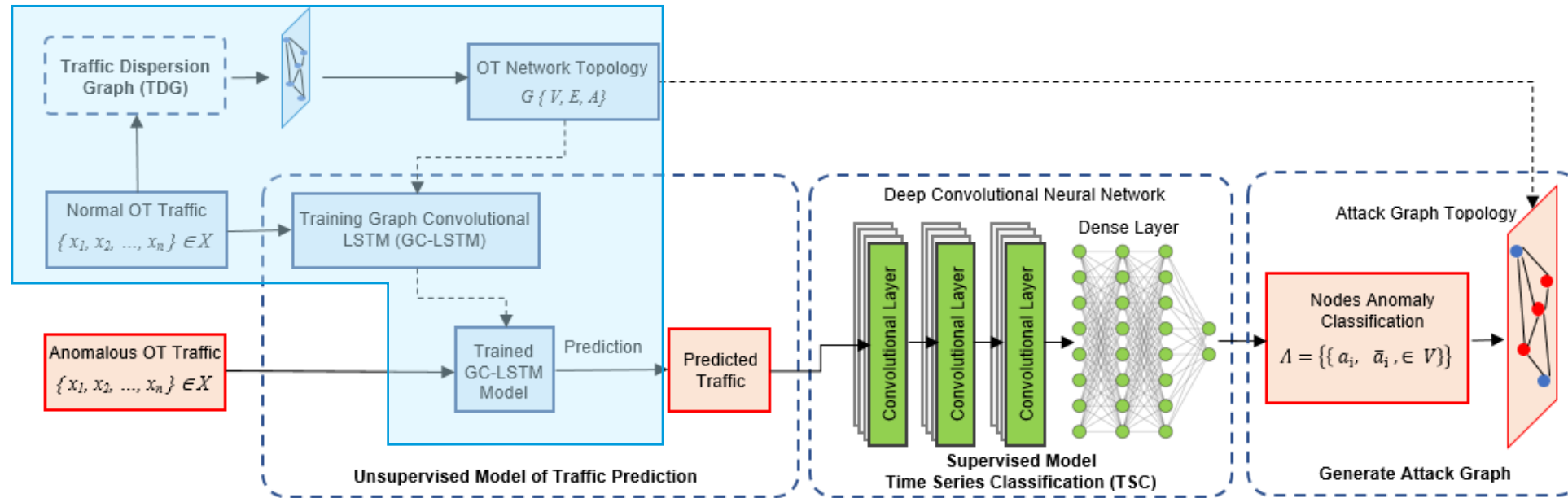
Deterministic Traffic

Anomaly Detection Strategies

- Hybrid deep learning model combines Graph Convolutional Long Short-Term Memory (GC-LSTM) and a deep convolutional network to detect OT network anomalies caused by cyber attacks
- GC-LSTM used to learn OT traffic behavior and generate normalization
- Traffic classification based on deep convolutional network with hyper parameters tuning through Bayesian optimization

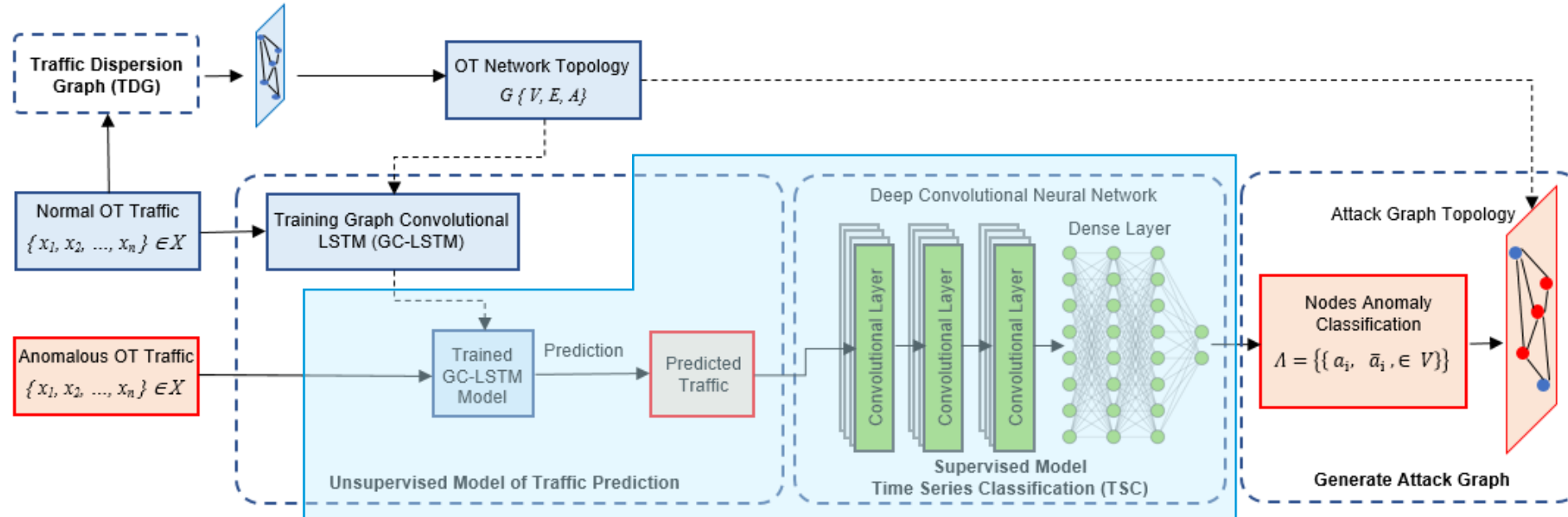


Attack Graph Creation Process (1)



- Step 1 - GC-LSTM training and Traffic Dispersion Graph (TDG)
 - The normal OT traffic is used to train the GC-LSTM model for traffic prediction. The process generates a trained GC-LSTM model
 - The normal OT traffic is used to generate the OT network topology using a TDG

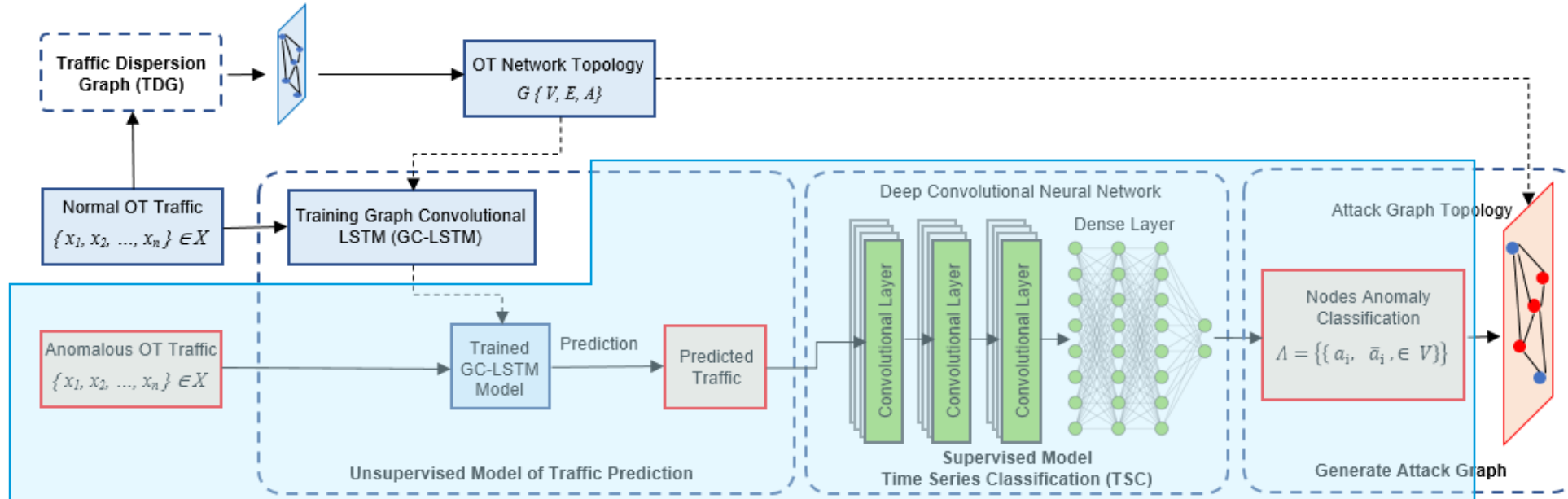
Attack Graph Creation Process (2)



- Step 2 - Deep Convolutional Neural Network (CNN) training

- The trained GC-LSTM model is used to predict the OT network traffic. The prediction is then used to train a Deep Convolutional Neural Network for TSC
- This process generates a trained Deep CNN model for OT traffic classification

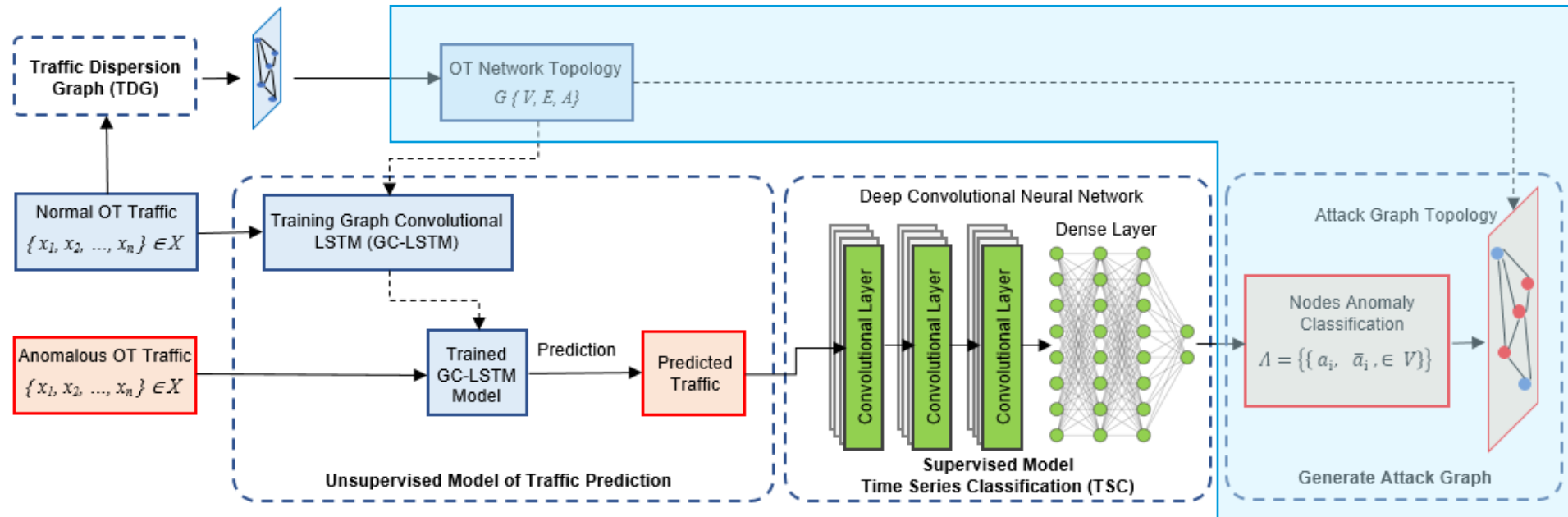
Attack Graph Creation Process (3)



- Step 3 - Online node classification

- This step monitors the online OT traffic as input for node classification.
- The trained GC-LSTM and Deep CNN are used sequentially to classify the nodes as normal or anomalous

Attack Graph Creation Process (4)

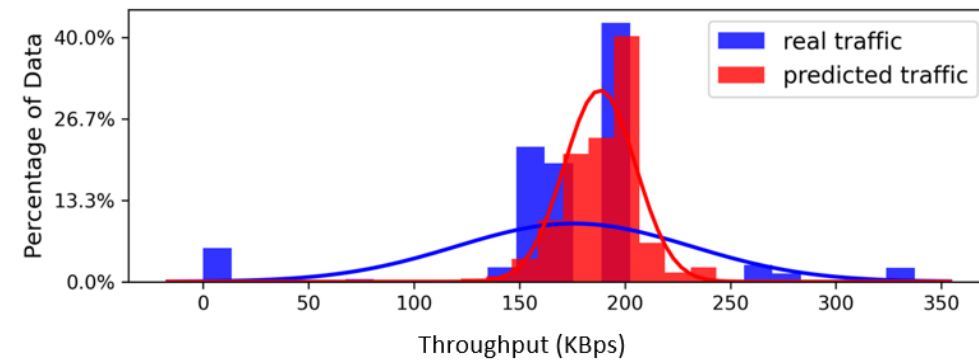
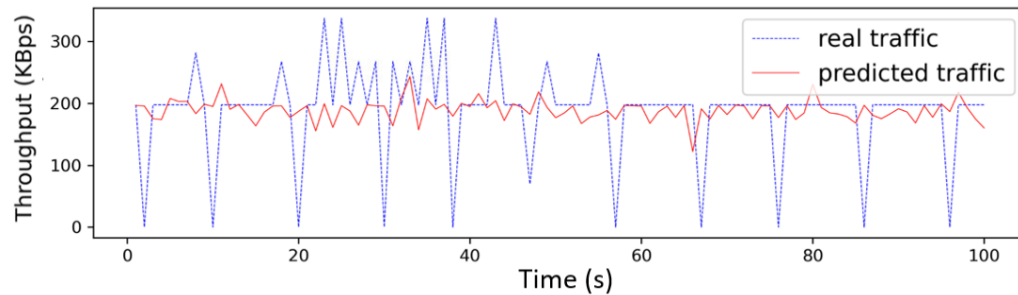


- Step 4 - Attack graph generation

- The node classification results from step 3 in conjunction with OT graph data from step 1 are used to generate the attack graph visualization

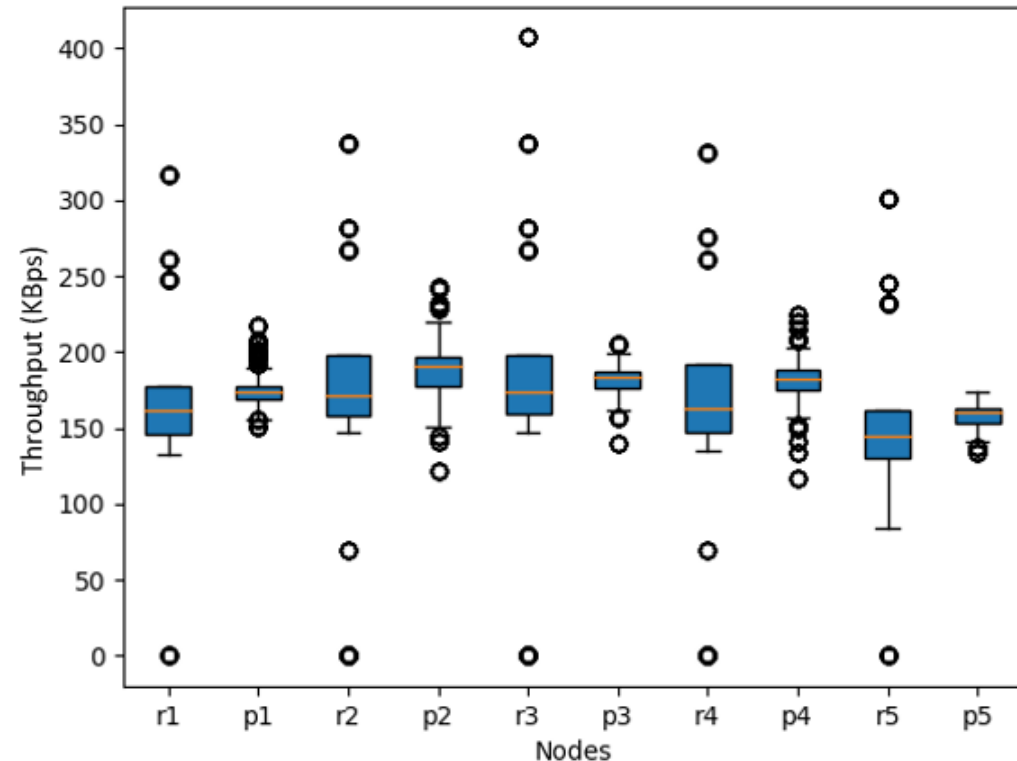
GC-LSTM for OT Traffic Normalization

- GC-LSTM learn spatial and temporal correlation of OT traffic through Graph Convolutional Network (GCN) and Long Short-Term Memory (LSTM)
- GC-LSTM produce predicted traffic as a filter to normalize and reduce the variability and outliers of OT traffic
- GC-LSTM reduces the false positive rate for the anomaly detection

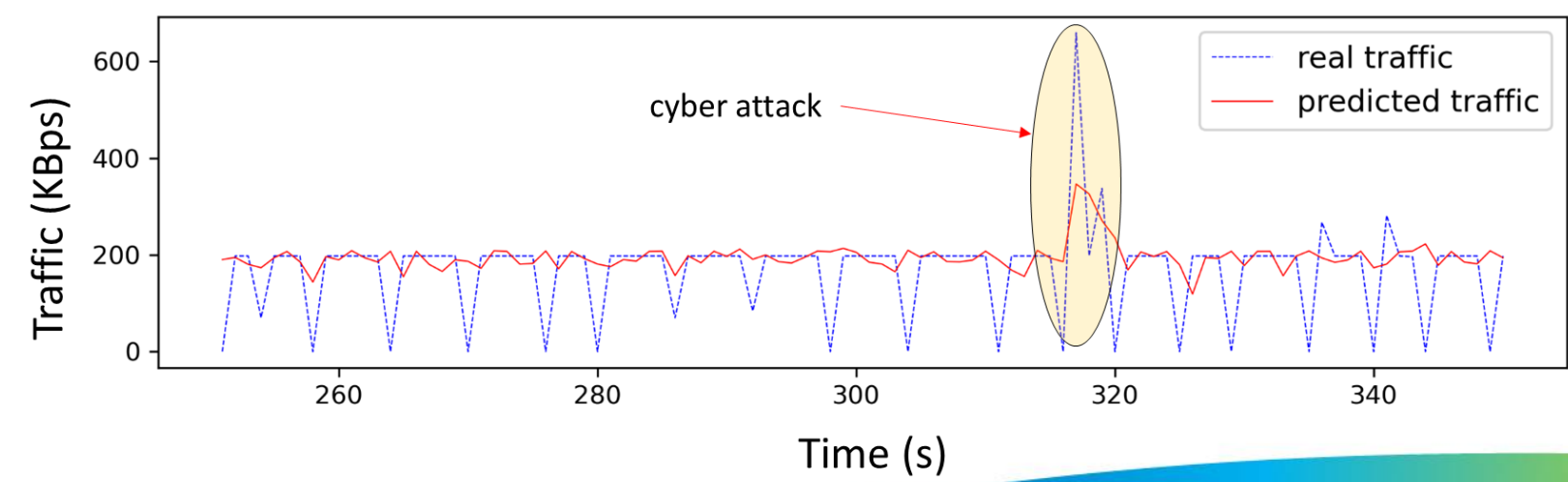
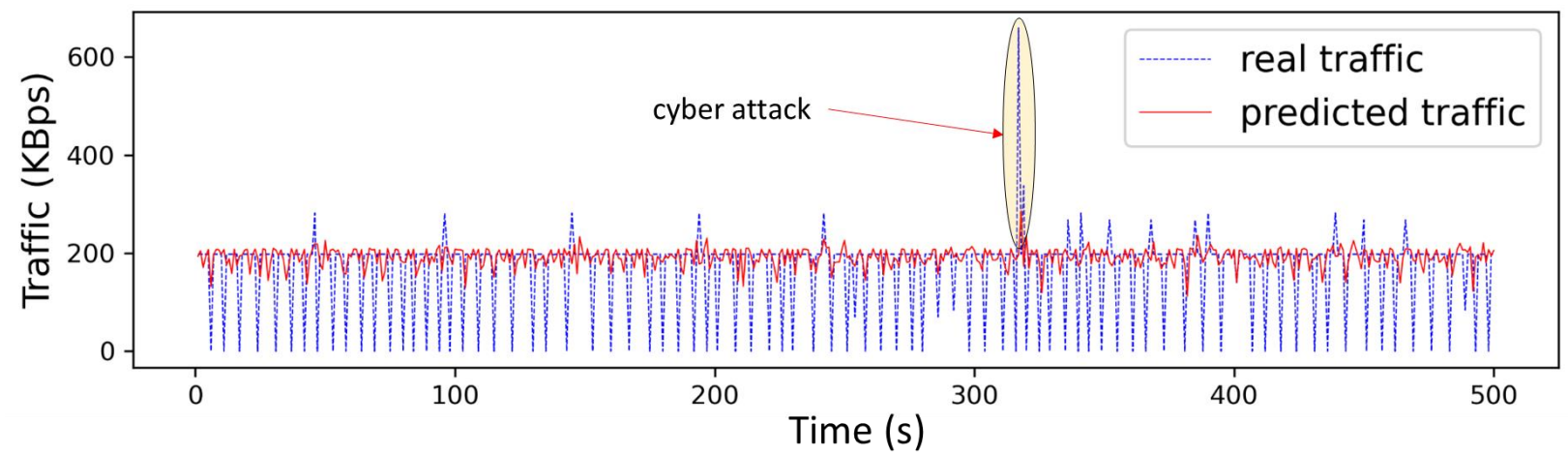


GC-LSTM for OT Traffic Normalization

- The figure below shows a statistical comparison of real (r) and predicted traffic (p) during normal operation.
- The predicted OT traffic has a more concentrated value and fewer outliers compared to the real data.

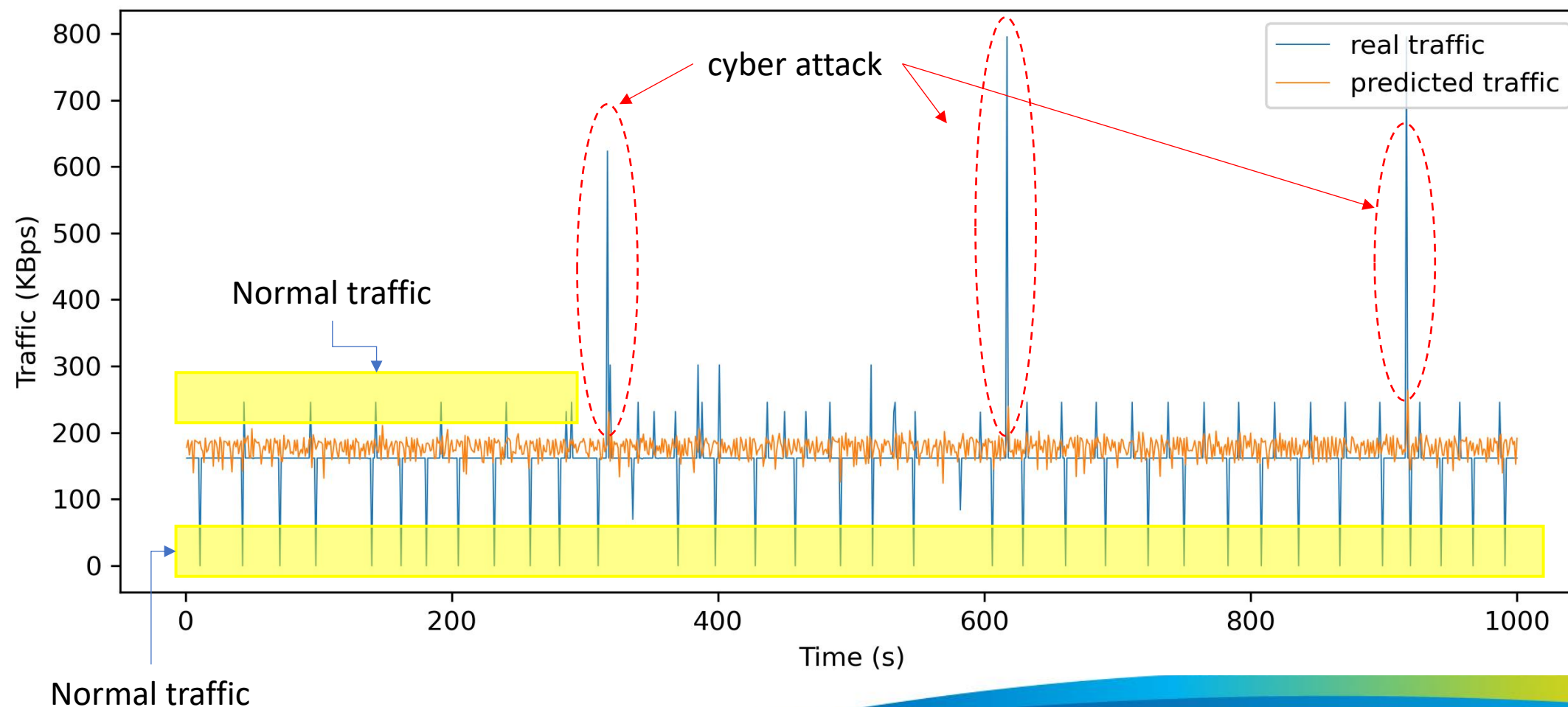


Normal vs Predicted Traffic Under Attack

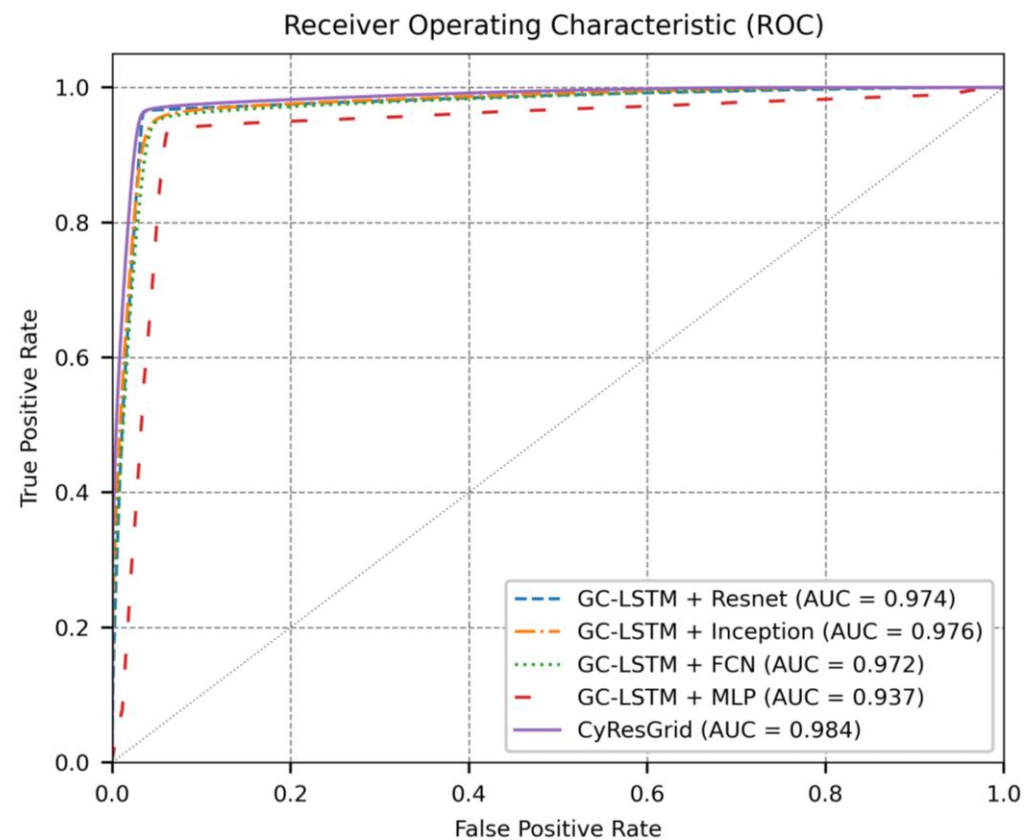
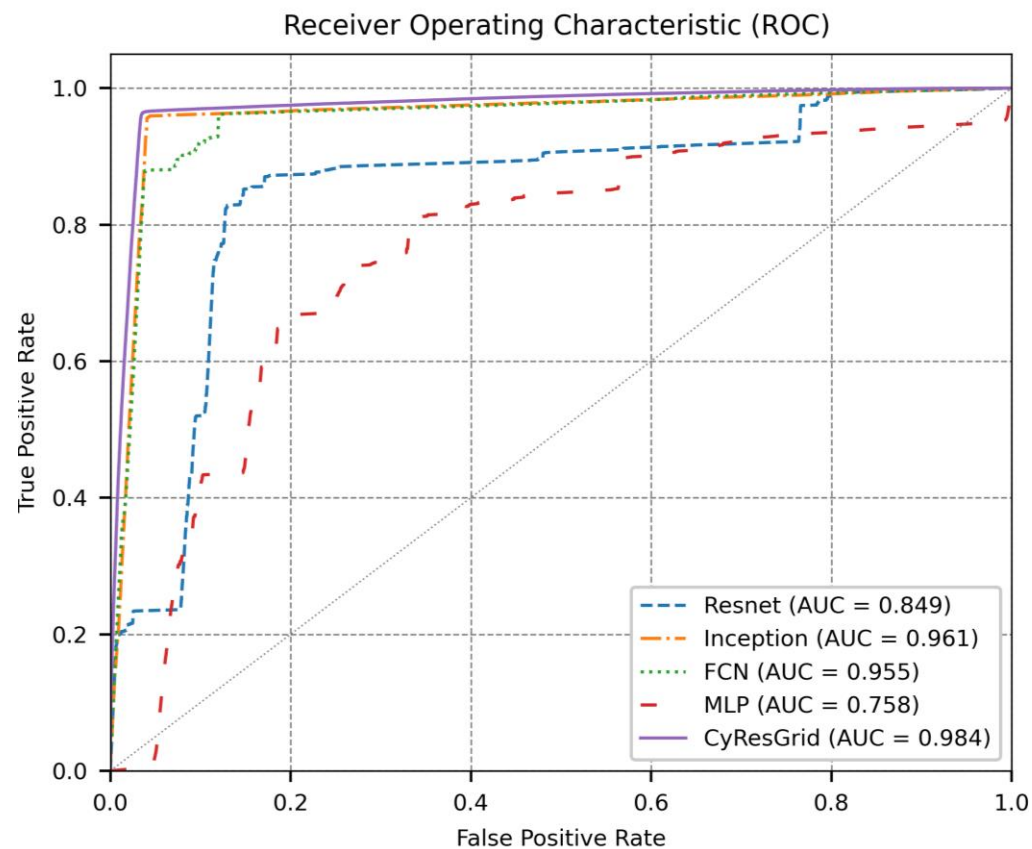


Reduce False Positive

- Predicted traffic from GC-LSTM produced a lower false positive rate.



Performance Comparison



Performance Comparison

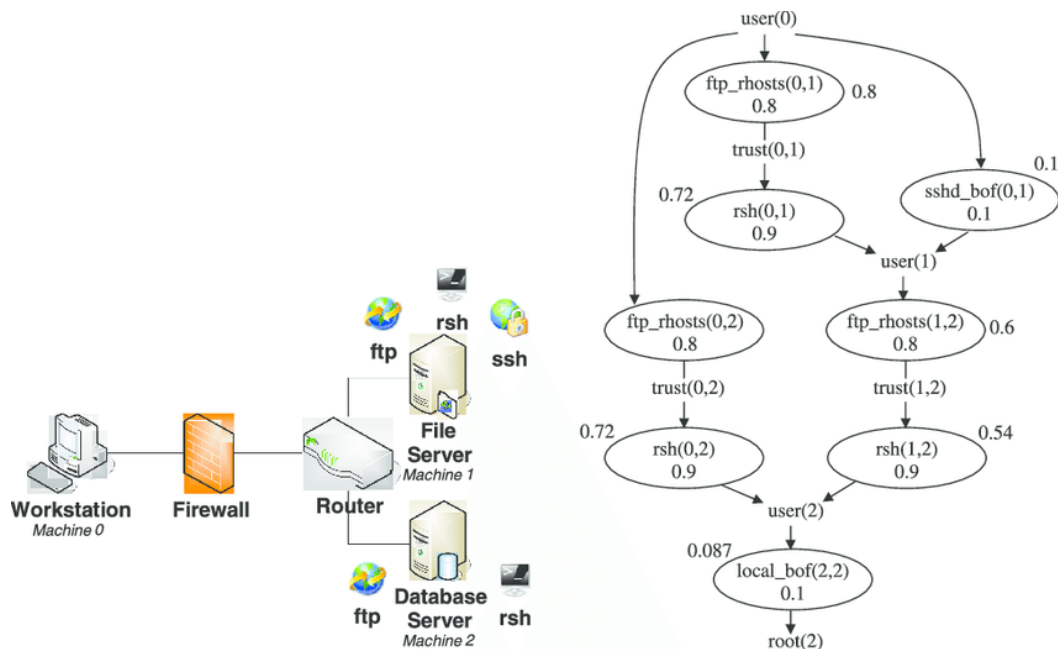
No	Methods	AUC	TN	FP	FN	TP	Accuracy	F1	G mean	Time (s)
<i>Combined attack scenarios</i>										
1	ResNet	0.849	82.27	11.32	3.49	2.92	85.19	28.29	15.50	633
2	Inception	0.961	93.50	0.20	4.10	2.31	95.71	51.76	14.68	976
3	FCN	0.955	88.16	5.43	3.92	2.49	90.65	34.76	14.81	1016
4	MLP	0.758	72.22	21.37	4.86	1.55	73.77	10.55	10.57	113
5	GC-LSTM + ResNet	0.974	93.29	0.31	3.27	3.14	96.42	63.77	17.12	1056
6	GC-LSTM + Inception	0.976	92.10	1.49	3.35	3.06	95.16	55.87	16.79	1409
7	GC-LSTM + FCN	0.972	92.28	1.30	3.68	2.73	95.01	52.26	15.87	1342
8	GC-LSTM + MLP	0.937	93.40	0.19	6.13	0.28	93.68	8.14	5.12	765
9	CyResGrid	0.984	93.47	0.13	3.42	2.99	96.45	65.03	17.16	714
<i>Stealthy attack scenarios</i>										
10	ResNet	0.8637	86.94	12.02	0.96	0.08	87.02	1.26	2.69	91
11	Inception	0.9887	98.93	0.02	1.04	0.0004	98.93	0.09	0.22	224
12	FCN	0.9833	87.82	11.13	1.01	0.02	87.85	0.47	1.58	240
13	GC-LSTM + ResNet	0.9524	89.93	9.02	0.95	0.09	90.02	1.87	2.92	226
14	GC-LSTM + Inception	0.9489	89.96	8.99	0.95	0.10	90.05	1.87	2.92	303
15	GC-LSTM + FCN	0.9491	89.96	8.99	0.95	0.10	90.05	1.87	2.92	304
16	CyResGrid	0.9243	91.15	7.81	0.94	0.111	91.25	2.32	3.08	138



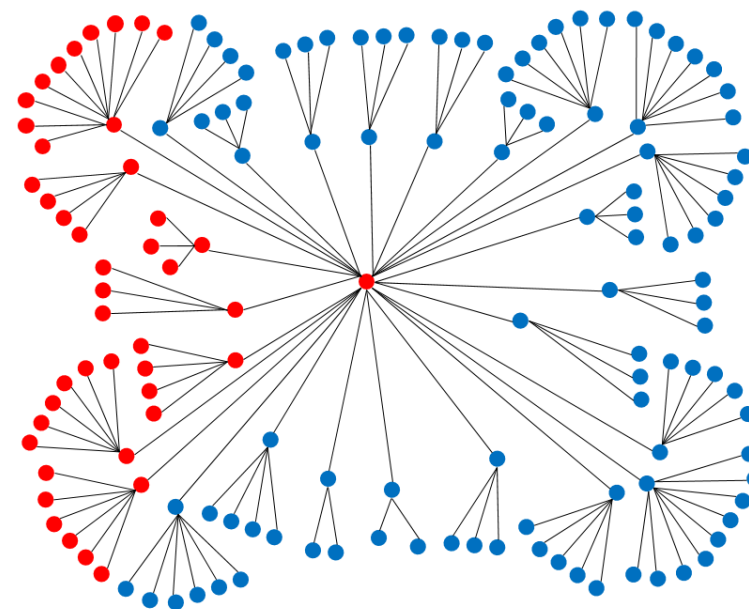
Attack Graph Model

Attack Graph

- An attack graph is a visual representation of the vulnerabilities and paths that attackers can exploit to compromise a system
- We propose an alternative attack graph map generation model, based on the online traffic monitoring in the OT networks of power grids



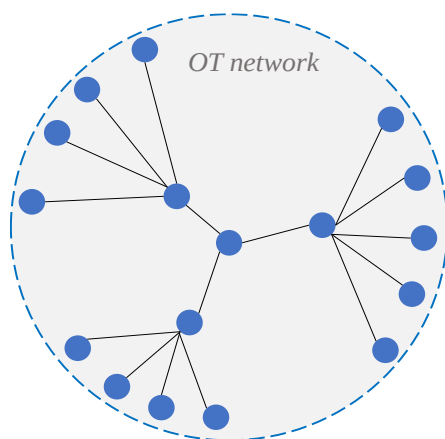
Typical Attack Graph



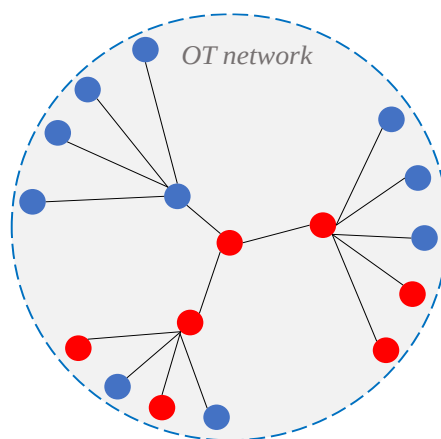
Proposed Attack Graph

Attack Graph Comparison

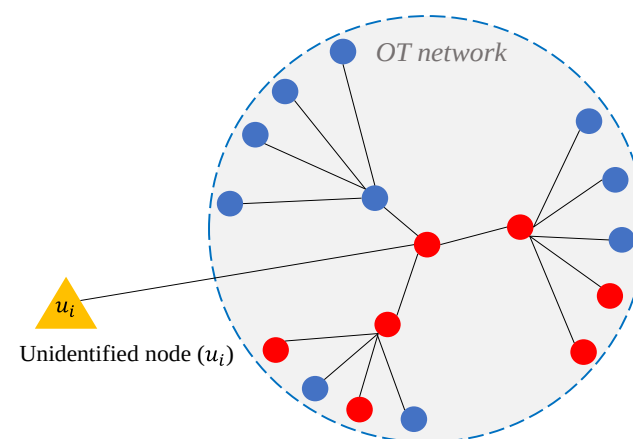
- Normal graph represent normal OT operation
- Attack graph type I is constructed based on prior knowledge of the OT network topology and node classification results.
- Attack graph type II considers unidentified nodes based on the TDG.



(a) Normal graph, $\{a_i, \in V\}$



(b) Attack graph type I, $\Lambda = \{a_i, \bar{a}_i, \in V\}$



(c) Attack graph type II, $\Lambda = \{\{a_i, \bar{a}_i, \in V\}, \{u_i \notin V\}\}$

V : known vertices/ nodes

Λ : attack graph

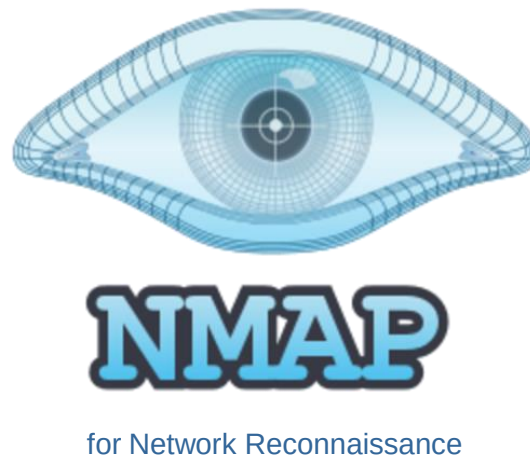
a_i : normal nodes

$\bar{a}_i$: anomalous nodes

u_i : unidentified nodes

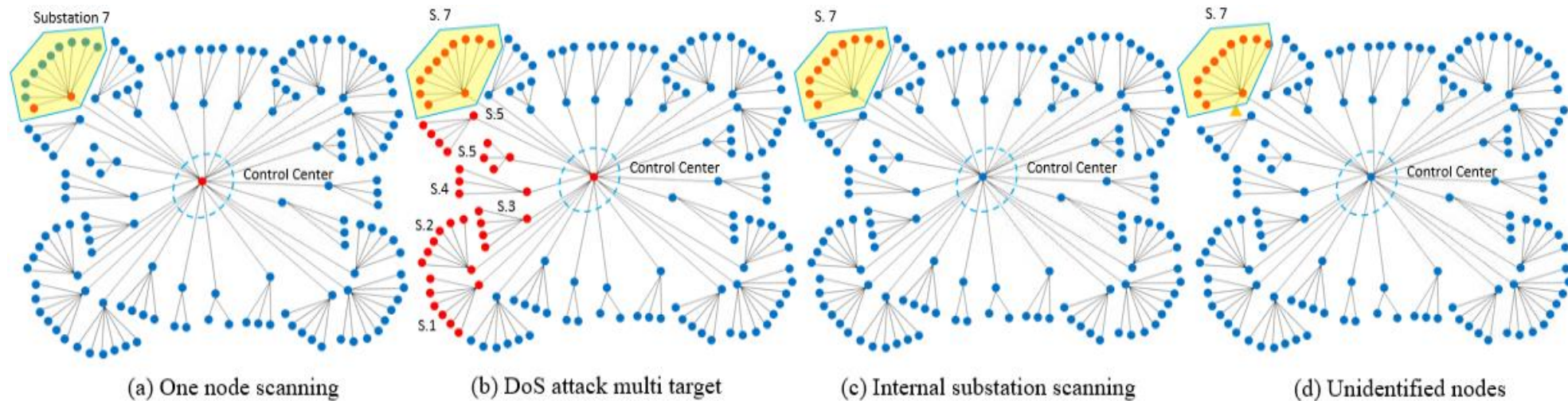
Cyber Attacks Experiments

- The cyber attacks run for 345,000 seconds and network throughput data is collected every second
- Simulated cyber attacks:
 - Denial of Service (DoS) with 3 level intensity: low, medium, and high
 - Network Reconnaissance with 6 level of intensity: paranoid, sneaky, polite, normal, aggressive and insane

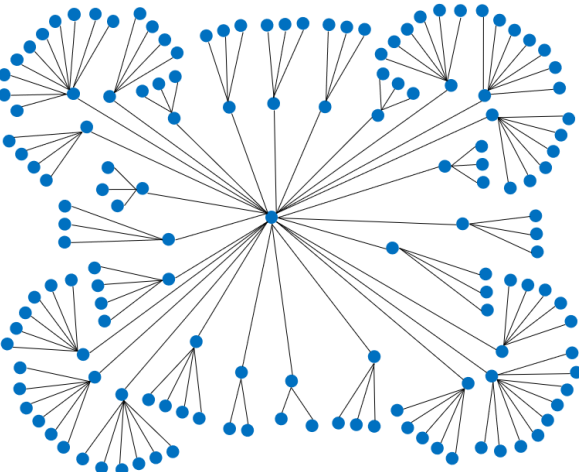


Attack Graph Maps to Identify and Visualize Cyber Attack Locations

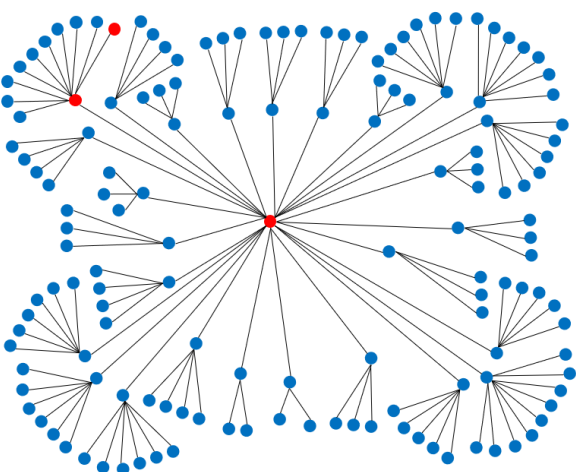
- Figure (a) depicts OT network scanning, originating from the control center to an OT device in substation 7
- Figure (b) depicts a DDoS attack targeting substations 1-7 that originates from the control center
- Figure (c) shows OT network scanning performed by a compromised OT device located in substation 7
- Figure (d) shows OT network scanning by an unidentified node, as indicated by an orange triangle



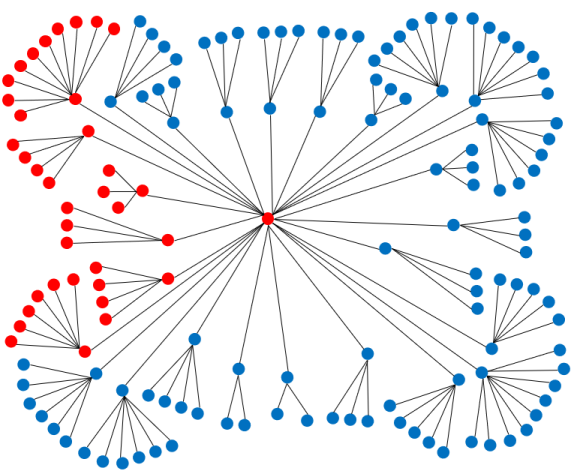
Attack Graph Visualization



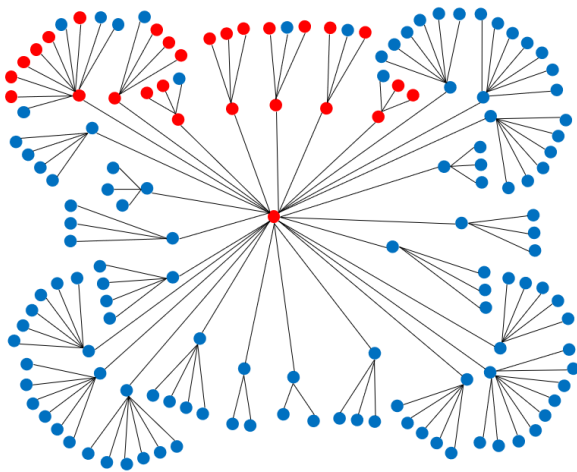
(a) Normal



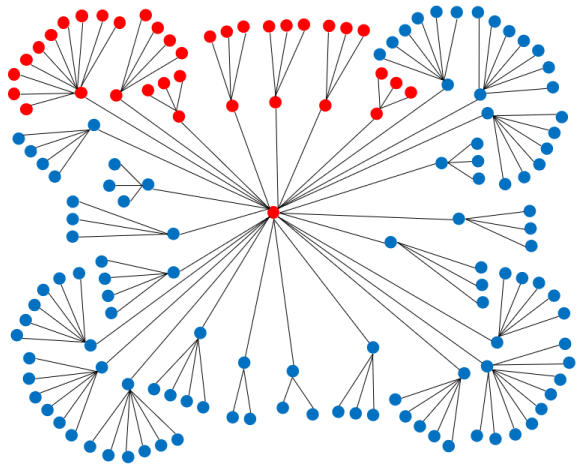
(b) DoS one target



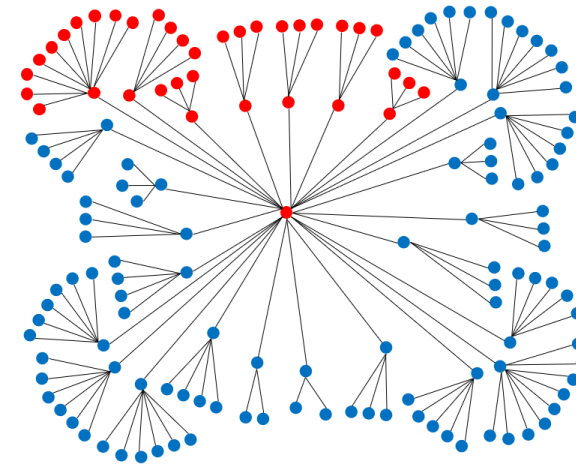
(c) DoS many targets



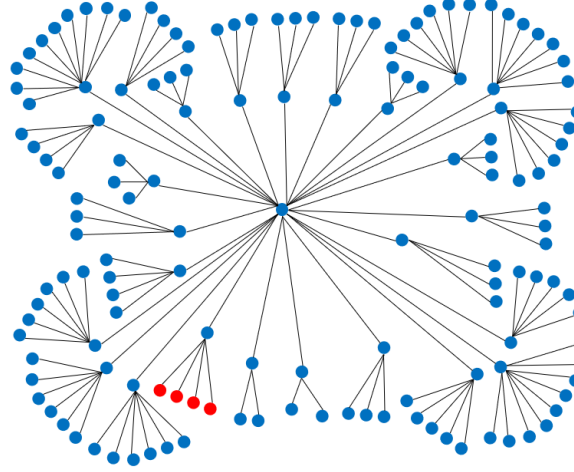
(d) Low intensity scanning



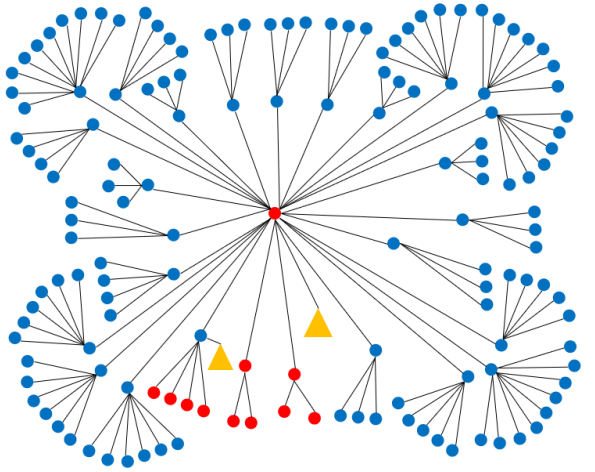
(e) Medium Intensity scanning



(f) High Intensity scanning

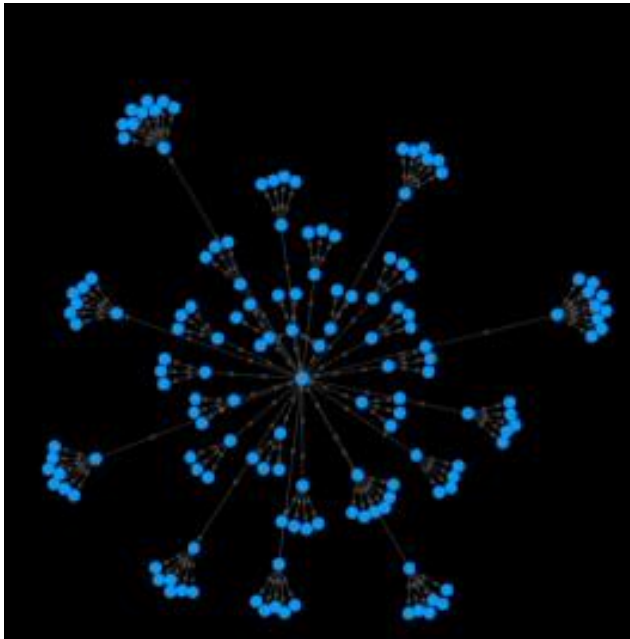


(g) Internal substation attack

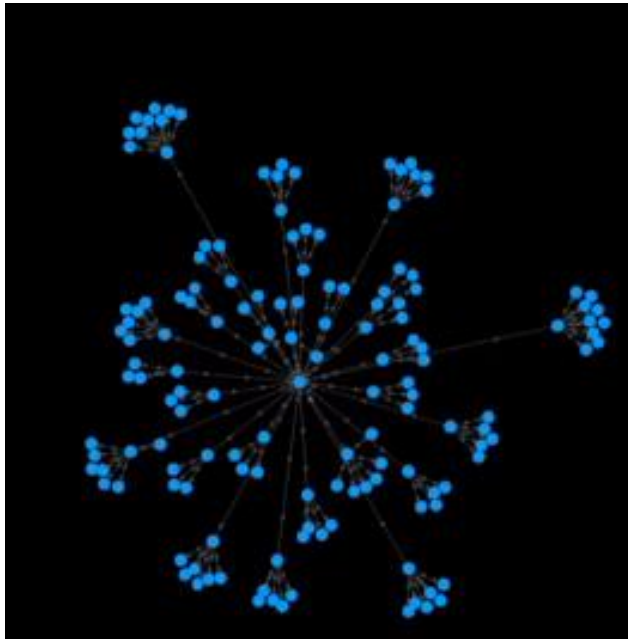


(h) Unidentified nodes

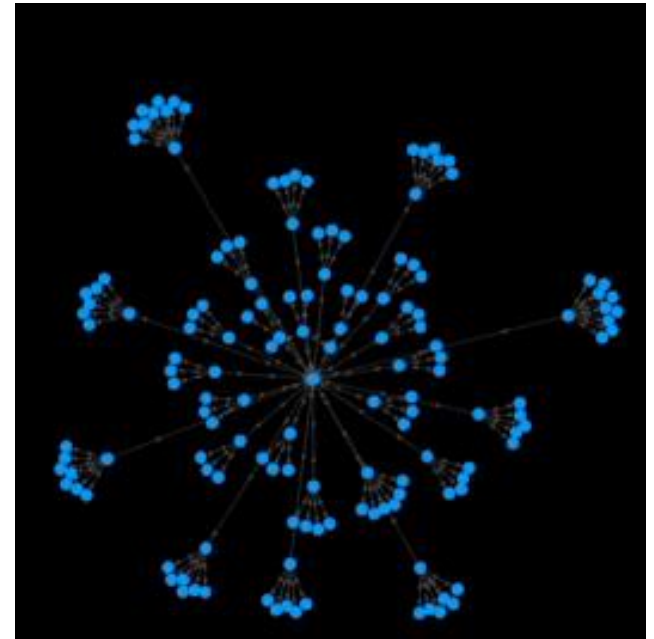
Attack Graph Demo



Single Substations Attack



Multiple Substations Attack

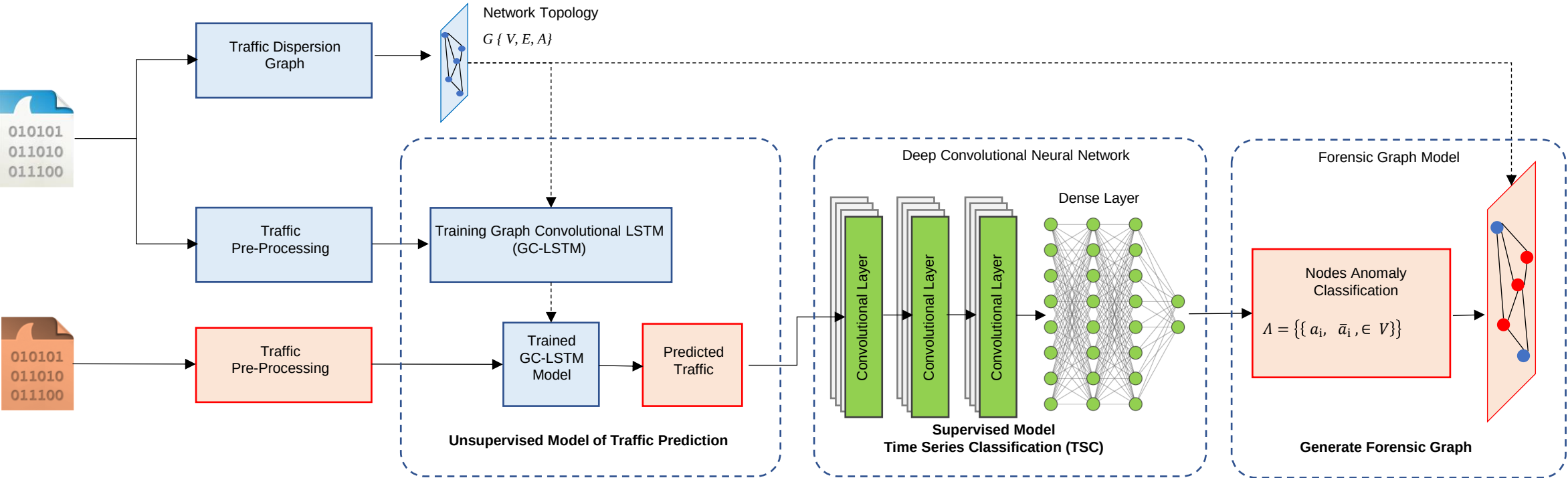


Wide Area Attack

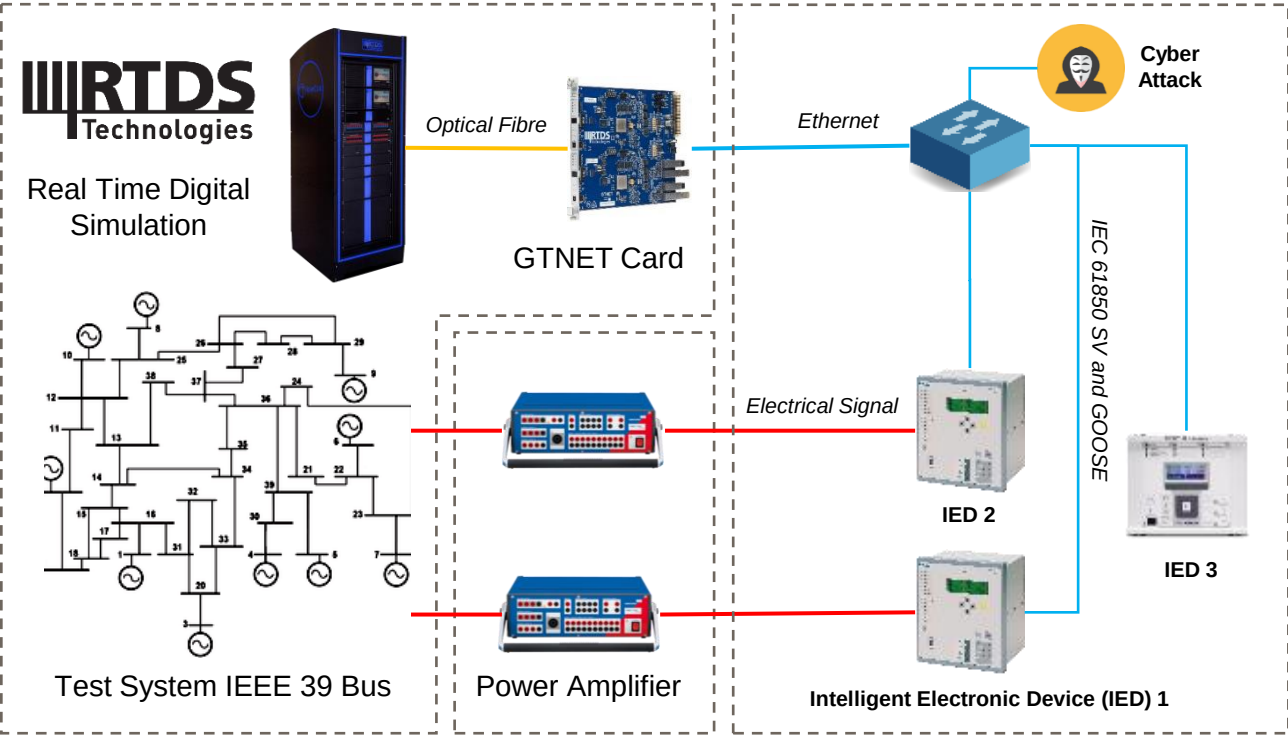
Conclusion

- CyResGrid uses hybrid model of GC-LSTM and a deep convolutional network for anomaly detection in OT communication networks for power grids.
- It helps power system operators to localize and identify cyber attacks in near real-time.
- GC-LSTM creates OT traffic predictions based on the spatial and temporal features of the input data. Through its predictions, the data variability and outliers are reduced. GC-LSTM also serves as a mechanism to improve the anomaly detection performance of TSCs.
- CyResGrid outperforms the state-of-the-art deep learning-based TSC.
- For stealthy cyber attack scenarios, i.e., paranoid and sneaky attacks, CyResGrid provides the best performance indicated by the highest F1 score of 2.32% and G mean score of 3.08%.

Forensic Graph Model



Hardware in the Loop Experimental Setup and Datasets



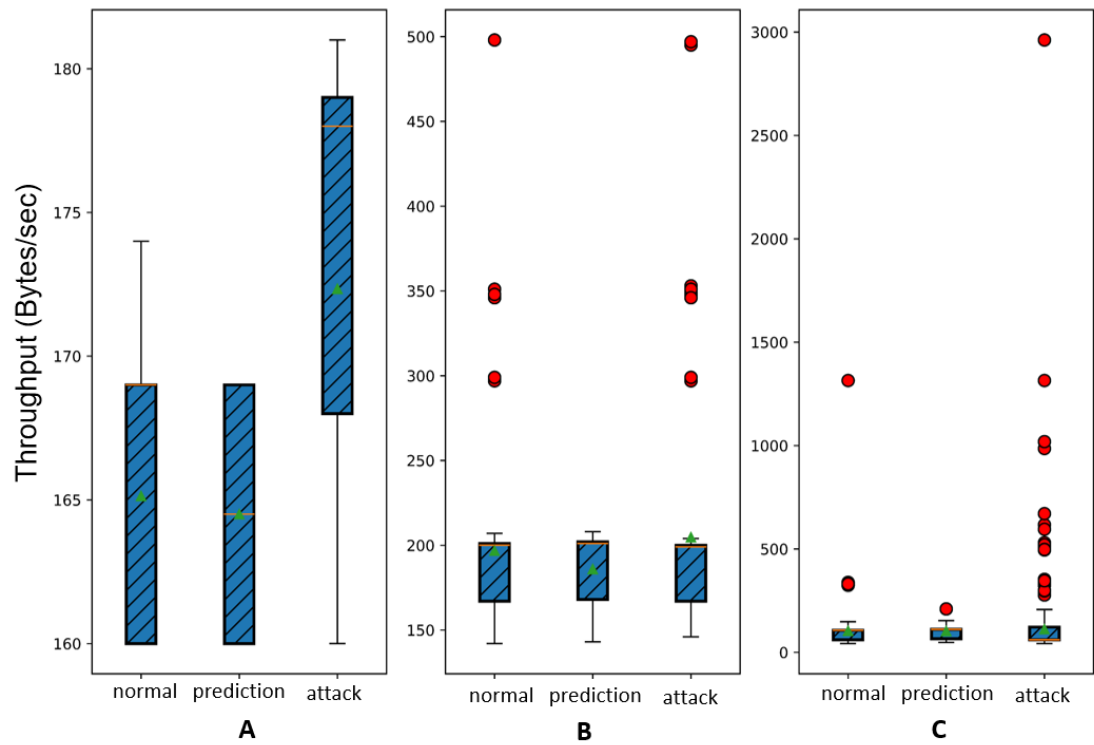
Parameters	A	B	C
No of Nodes	85	103	786
No of Edges	198	246	821
Traffic duration	30 minutes	150 minutes	5 days
Total packet size	50 MB	100 MB	17 GB

- A: HIL Experiment Data
- B: IEC61850 Dataset
- C: DAPT Dataset

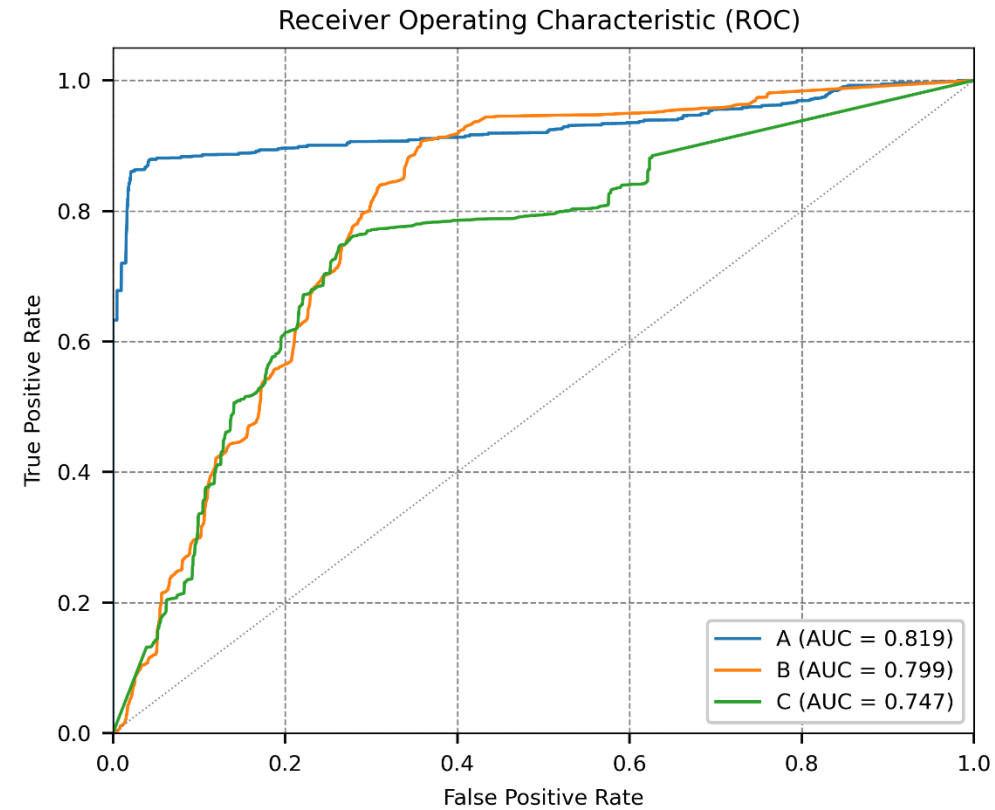
[B] P. P. Biswas, H. C. Tan, Q. Zhu, Y. Li, D. Mashima and B. Chen, "A Synthesized Dataset for Cybersecurity Study of IEC 61850 based Substation," *IEEE SmartGridComm*, Beijing, China, 2019, pp. 1-7.

[C] S. Myneni, A. Chowdhary, A. Sabur, S. Sengupta, G. Agrawal, D. Huang, et al., "DAPT 2020—Constructing a benchmark dataset for advanced persistent threats" in *Deployable Machine Learning for Security Defense*, Cham, Switzerland:Springer Int, pp. 138-163, 2020.

Experimental Results Forensic Graph

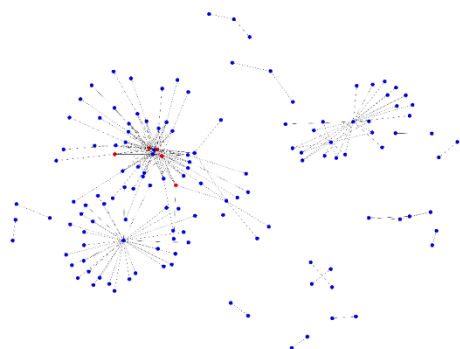


Statistical Comparison of Traffic Data

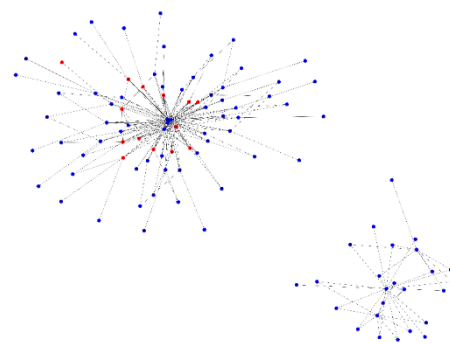


Classification Performance

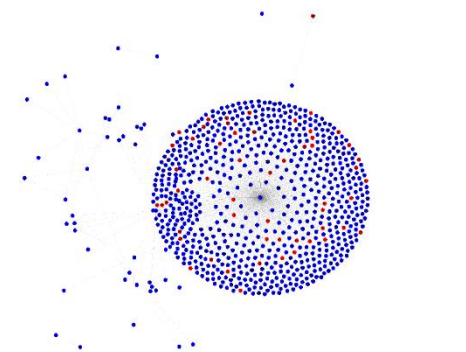
Forensic Graph Results



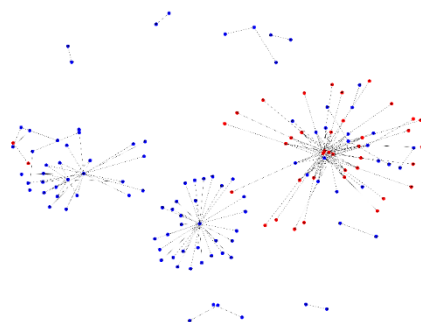
a) A under GOOSE Attack



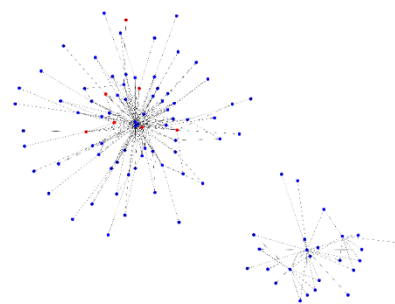
b) B under GOOSE Attack



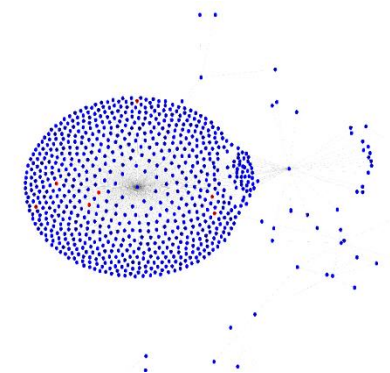
c) C under Reconnaissance Attack



d) A under Reconnaissance Attack



e) B under Data Manipulation Attack

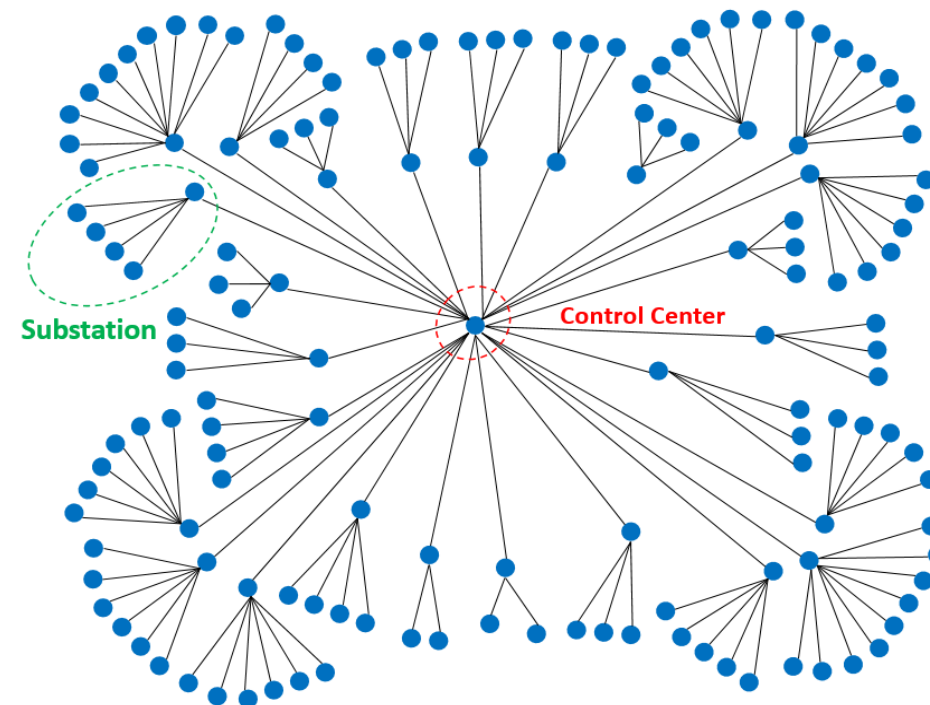
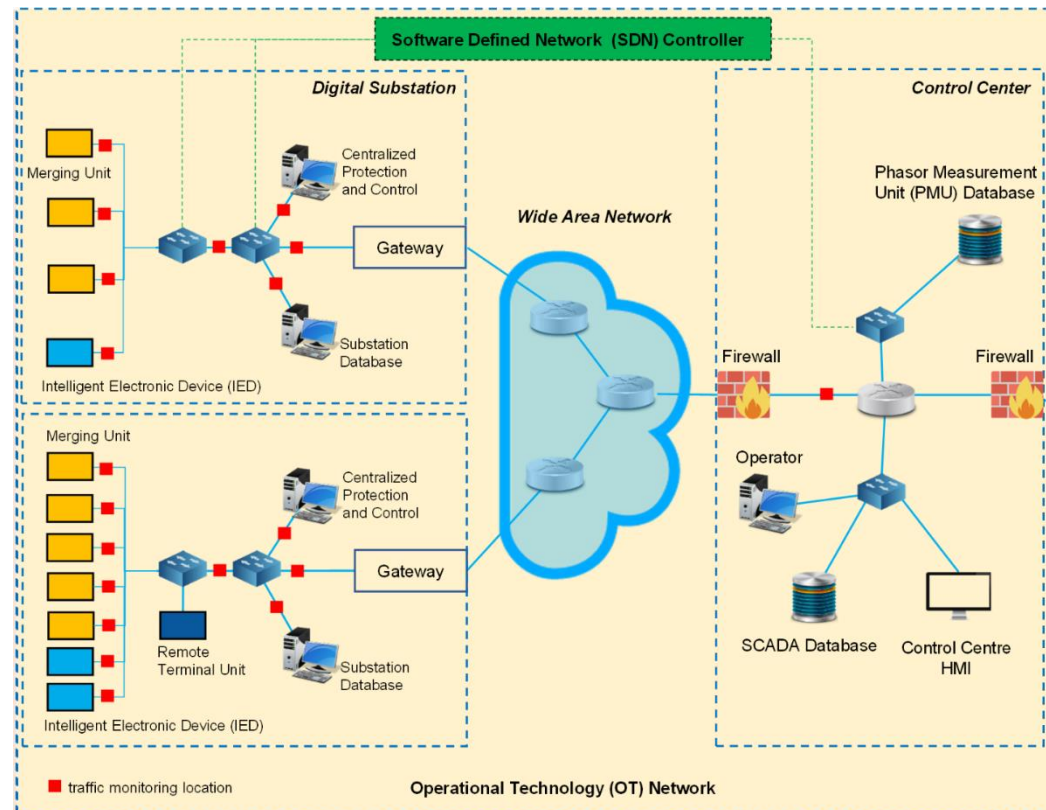


f) C Establish Foothold Attack



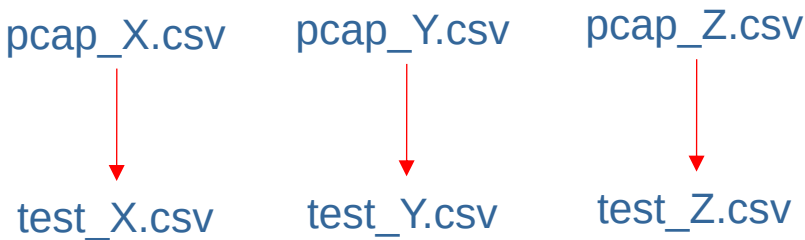
Lab 5: Anomaly Detection and Time Series Classification

Simulated Network



Dataset (1)

No.	Time	Source	Destination	Protocol	Length	Info
1	0	100.7.0.16	100.0.0.11	TCP	116	991 > 41
2	2E-05	100.0.0.11	100.7.0.16	TCP	66	41610 >
3	0.057	100.7.0.15	100.0.0.11	TCP	96	991 > 40
4	0.057	100.0.0.11	100.7.0.15	TCP	66	40760 >
5	0.1081	100.7.0.18	100.0.0.11	TCP	124	991 > 33



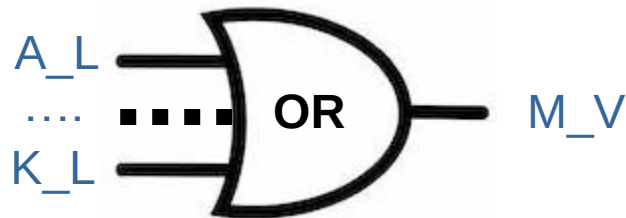
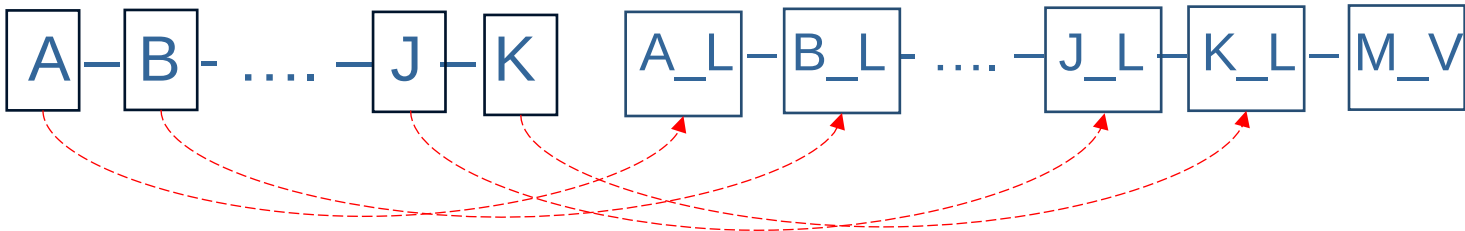
A	B	C	D	E	F	G	H	I	J	K
176	198	198	192	162	182	172	190	162	190	1822
176	198	198	0	162	182	172	190	162	190	1630
176	198	0	192	162	182	172	190	162	190	1624
176	198	198	192	162	182	172	190	162	0	1632

Time	100.0.0.11- 100.7.0.11	100.0.0.11- 100.7.0.12	100.0.0.11- 100.7.0.13	100.0.0.11- 100.7.0.14	100.0.0.11- 100.7.0.15
0.0	176	198.0	198.0	192	162
1.0	176	198.0	198.0	0	162

Dataset (2)

A	B	C	D	E	F	G	H	I	J	K	A_L	B_L	C_L	D_L	E_L	F_L	G_L	H_L	I_L	J_L	K_L	M_V
173.1459	196.2947	189.9267	190.4694	161.2178	184.6467	176.3418	187.241	170.0572	168.8464	1616.542	0	0	0	0	0	0	0	0	0	0	0	0
169.751	177.1935	171.6637	170.0589	162.184	176.7735	158.5844	190.6844	153.4272	164.1016	1595.559	0	0	0	0	0	0	0	0	0	0	0	0
1256.144	1993.315	1817.861	1807.415	2277.384	2153.864	2237.028	2654.627	1628.393	1242.824	17744.84	1	0	0	0	0	0	0	0	0	0	1	1
169.9071	185.2044	181.247	177.8686	167.7897	178.5435	162.9958	211.2225	161.0363	167.9885	1610.13	0	0	0	0	0	0	0	0	0	0	0	0

tsc_training_data.csv



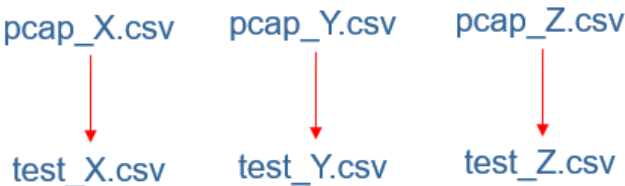
Assignment Part 1

1. Network Traffic Analysis Using Throughput and TDG.

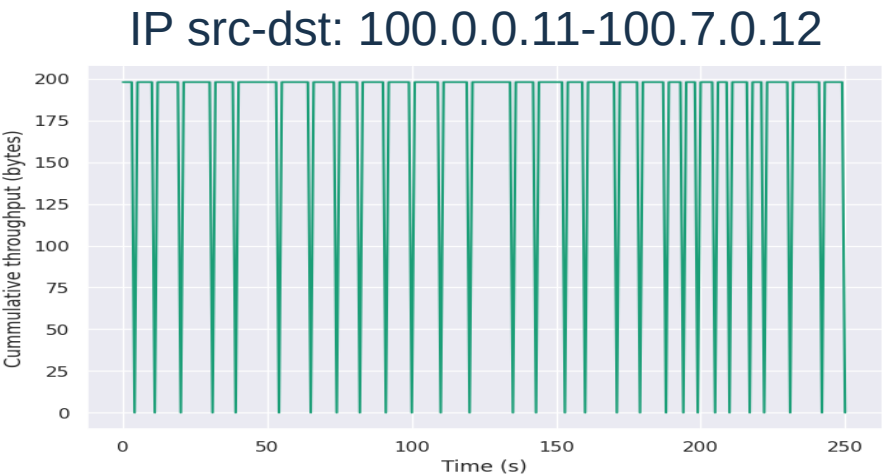
Amongst the 3 scenarios (pcap_X.csv, pcap_Y.csv, and pcap_Z.csv):

- a. Which of them are classified as normal traffic and cyber attack traffic? Support your answer with data analysis and plots.
- b. Identify what are the adversary IP address and attack target IP addresses for each scenario.

No.	Time	Source	Destination	Protocol	Length	Info
1	0	100.7.0.16	100.0.0.11	TCP	116	991 > 41
2	2E-05	100.0.0.11	100.7.0.16	TCP	66	41610 >
3	0.057	100.7.0.15	100.0.0.11	TCP	96	991 > 40
4	0.057	100.0.0.11	100.7.0.15	TCP	66	40760 >
5	0.1081	100.7.0.18	100.0.0.11	TCP	124	991 > 33



A	B	C	D	E	F	G	H	I	J	K
176	198	198	192	162	182	172	190	162	190	1822
176	198	198	0	162	182	172	190	162	190	1630
176	198	0	192	162	182	172	190	162	190	1624
176	198	198	192	162	182	172	190	162	0	1632



Time	100.0.0.11-100.7.0.11	100.0.0.11-100.7.0.12	100.0.0.11-100.7.0.13	100.0.0.11-100.7.0.14	100.0.0.11-100.7.0.15
0.0	176	198.0	198.0	192	162
1.0	176	198.0	198.0	0	162

Assignment Part 2

2. Time Series Classification using Machine Learning.

Use the labelled dataset `tsc_training_data.csv` to train a TSC machine learning model.

Note: You are allowed to test and modify the model and hyperparameters to improve classification performance!

Implement the trained machine learning model to classify unlabelled time series data of `test_X.csv`, `test_Y.csv`, and `test_Z.csv`. Based on the prediction results, answer the following questions:

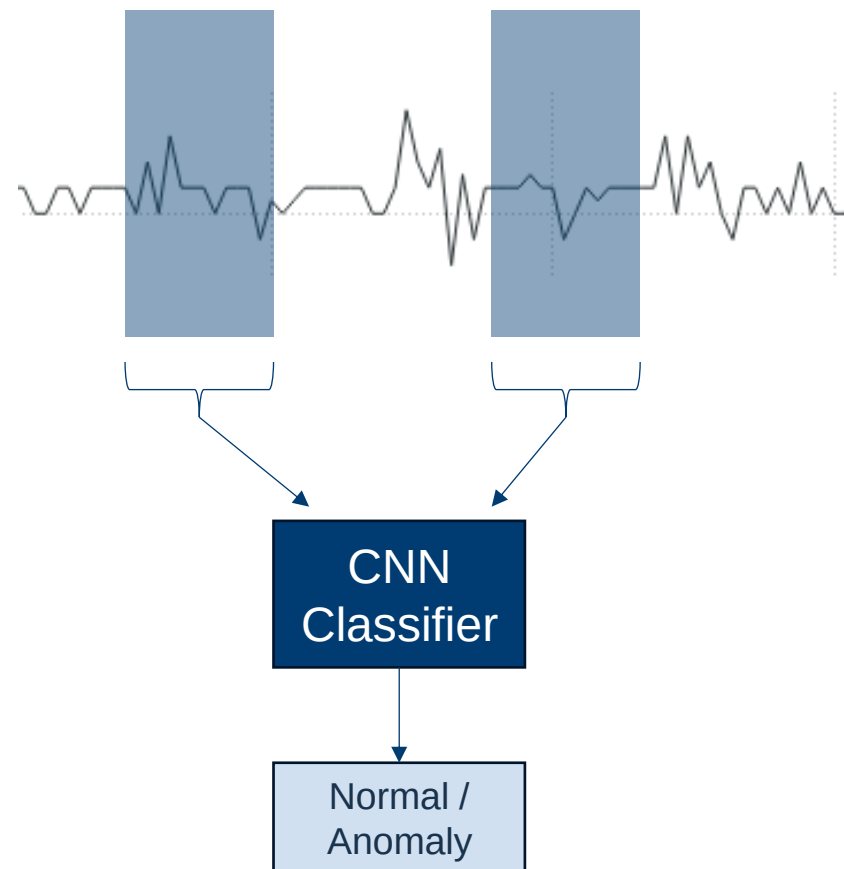
- Among all test data, which of them are classified as normal traffic and cyber attack traffic?
- Identify the impact of cyber attack on the time series traffic. Support your answer with data analysis and plots.

```
### Time Series Univariate Classification
selected_data = 'A'
selected_label = 'A_L'
n_classes = 2
sequence_length = 4

X, y = prepare_train_dataset_univariate(train_data, selected_data, selected_label, n_classes, sequence_length)
X_test = prepare_test_dataset_univariate(test_data, selected_data, sequence_length)
```

```
# Define the model
model = Sequential([
    Conv1D(filters=64, kernel_size=3, activation='relu', input_shape=(sequence_length, 1)),
    MaxPooling1D(pool_size=2),
    Flatten(),
    Dense(50, activation='relu'),
    Dense(n_classes, activation='softmax')
])

# Compile the model
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
```



Assignments

3. Provide a comparative analysis of the two methods (TDG and TSC) used in Parts 1 and 2. List the advantages and drawbacks for each method.
4. Based on the lectures and assignment, provide suggestions/ideas on how to detect anomalous traffic in power grid operational technology networks?



GET IN
TOUCH

Alex Stefanov

Assistant Professor, Chartered Engineer (CEng MIEI)

Email: A.I.Stefanov@tudelft.nl

Cyber Resilient Power Grids ([LinkedIn](#))

Control Room of the Future ([LinkedIn](#))

Intelligent Electrical Power Grids, EEMCS, **TU Delft**
Mekelweg 4, 2628 CD Delft, The Netherlands